SORA

Sobaie Rainfall Analysis

User Manual and Technical Guide

Rainfall frequency analysis, design storms and depth–/intensity–duration–frequency (DDF/IDF) relationships — from raw gauge records to a defensible engineering report. Every input, calculation and output is explained together with the standard or publication it is based on.

Tool	SORA – Sobaie Rainfall Analysis, Windows desktop app (portable "SORA 1.1.exe")
Author	Mohamed Abosobaie (Sobaie)
Document	User manual, methods and references
Edition	November 2026 – for SORA version 1.1

How to read this manual: a R7 tag after a statement points to the reference list in Chapter 10; a Tool tag marks a convention chosen in SORA rather than taken from a standard. Pink numbered circles 1 on the screenshots match the numbered explanations next to them.

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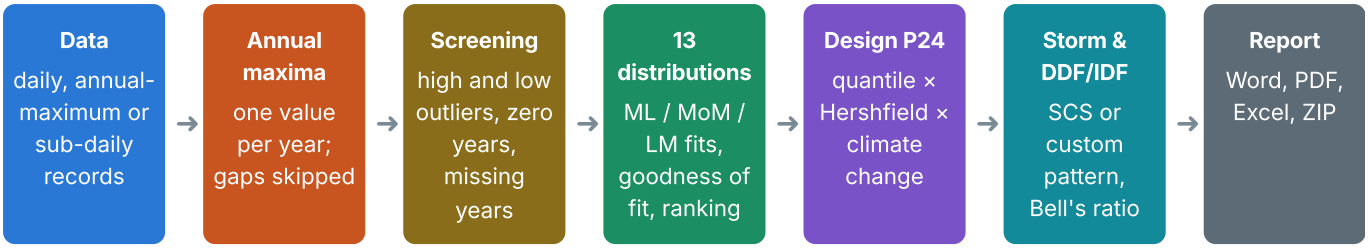
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Quick start (5 minutes)

1. Open step **1 Rainfall data** and drop your Excel, CSV or rain-gauge .txt file(s) into the box (or paste cells from Excel). Not sure of the layout? **Download templates (.xlsx)**. To try SORA first, click **Load example**.
2. Check step **2 Settings**: Hershfield factor (1.13 for daily data, 1.00 for sub-daily data), the climate change factor required by your project (**default 1.20** – set 1.00 for no allowance), the zeros & outliers option and the SCS storm type and duration.
3. Step **3 Distribution** and step **4 Return periods & tables** work with their defaults; change them only when your project asks for it.
4. Look at the **Summary** and **Annual maxima** tabs (outliers, zero and missing years, checks).
5. Click **Export ▾**, fill in the report cover (Prepared by, Checked by ...) and **Download** the PDF or Word report.
6. **Save** (top bar) keeps the work as a *.sora* project file – SORA starts empty each time.

1 What SORA does

SORA turns a rainfall record into design rainfall. It follows the standard single-station frequency-analysis chain used in hydrology R5 R21 R18:



1.1 One complete view

The four input steps sit in the left panel, each with an *Explanation of the inputs* at its foot; the results tabs fill the rest of the window. Results and reports are always the full, detailed version. For each distribution the best of its available estimation methods (ML, MoM, LM; Bulletin 17C moments for Log-Pearson III) is chosen automatically, unless a method is set in step 3 or on the Fitting tab; its abbreviation is shown next to the distribution name.

1.2 Running SORA

- **Portable desktop app** – the single file *SORA 1.1.exe*, from the SORA download page. No installation and no administrator rights: copy it anywhere (Desktop, USB stick) and double-click. It needs Microsoft WebView2, which is part of Windows 10 and 11.
- **Browser download warning** – SORA is not code-signed, so Chrome, Edge or Windows SmartScreen may warn about a new file. In Chrome open *Downloads*, then *Download dangerous file* / *Keep*; in Windows choose *More info* → *Run anyway*. A company may block such files – ask your IT department.
- **Internet at start** – SORA checks its access online when it starts (Chapter 2.1); the analysis itself runs on your computer.
- **Save as** – every export, project save and the user manual opens a Windows *Save as* window to choose the folder and file name.

Where the data are kept

All calculations run on your own computer; your rainfall data, projects and results are never uploaded. SORA starts with no gauges each time; keep your work as project files (*.sora*, Chapter 8.2). When you close SORA with changes not yet saved to a project file, it asks **Save / Don't save / Cancel**. Tool

Anonymous usage counts

As stated in the terms of use, SORA sends short anonymous counts so the author can see how widely it is used: that SORA was started, the version, the number of gauges analysed, the exports used, the country and a random number that identifies the installation (not the person). No rainfall data, results, file, gauge or project names or personal details are ever sent; when the computer is offline nothing is sent. Tool

2 Getting started – access and the screen

2.1 Opening SORA – access and messages

SORA checks its access online every time it starts, so an **internet connection is needed at start**; it does not open offline. What you see depends on how the developer has set access:

- **Open access** – SORA opens straight away, no password.
- **Password** – type the password you were given and press **Open SORA**. It is remembered on this computer for **14 days**: until then SORA opens without asking. It is asked again after 14 days, when the developer changes the password, or when open access is switched off.

While SORA is open it re-checks every few hours; if access has ended you are asked to save your work, then SORA closes. The start can also show:

- **Messages** from the developer – notices, and popups shown once per computer (several unseen ones together). **OK** becomes active after a short 3-second ring, so the message is read.
- **New version of SORA** – a message with **Later** and **Download**: Download saves the new SORA in your Downloads folder, shows the progress and opens it; the old copy stays until you delete it. When the developer requires a newer version, an older copy does not open and offers only Download.
- **Too many attempts** (three wrong passwords within 15 minutes), **Daily limit for this computer reached**, or **This computer has been blocked** – wait, or contact the developer (mohamedabosebie@gmail.com).

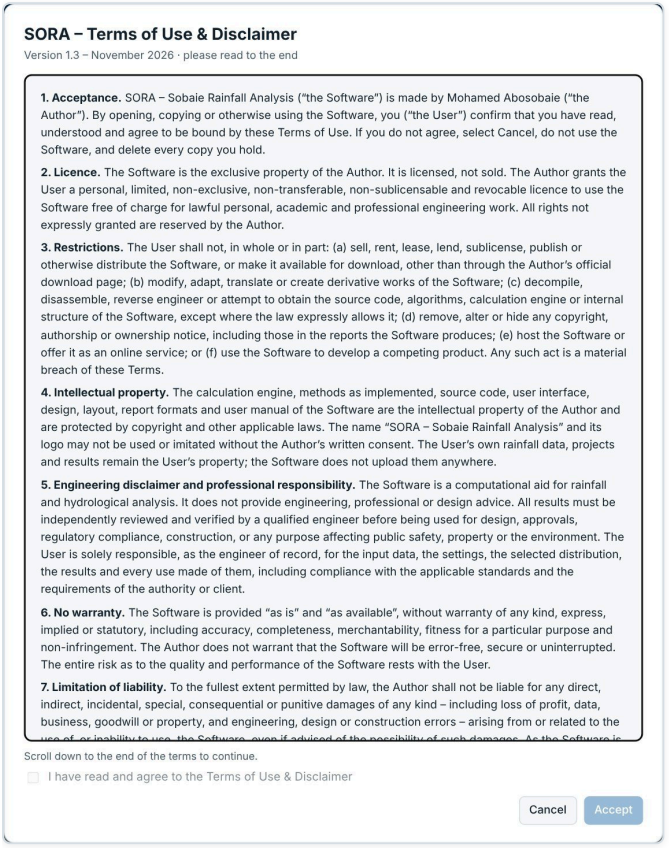


Figure 2.1 – Terms of use at the first start

Opening for the first time

At the first start SORA shows its **Terms of Use & Disclaimer**. Scroll to the end of the text, tick *I have read and agree…* and click **Accept**; **Cancel** closes SORA. The terms are shown again only when a new version of them is issued, and can be read at any time from **Terms of use** in the top bar. Each computer (installation) accepts them once.

SORA then opens with one empty gauge: load data in step 1, open a project, or click **Load example**.

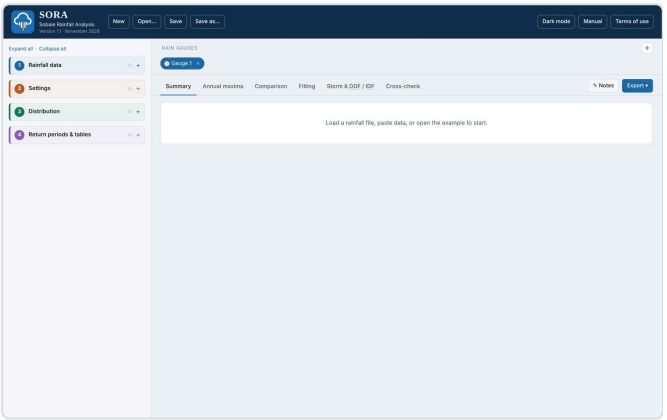


Figure 2.2 – SORA after the terms are accepted (no data yet)

2.2 The screen

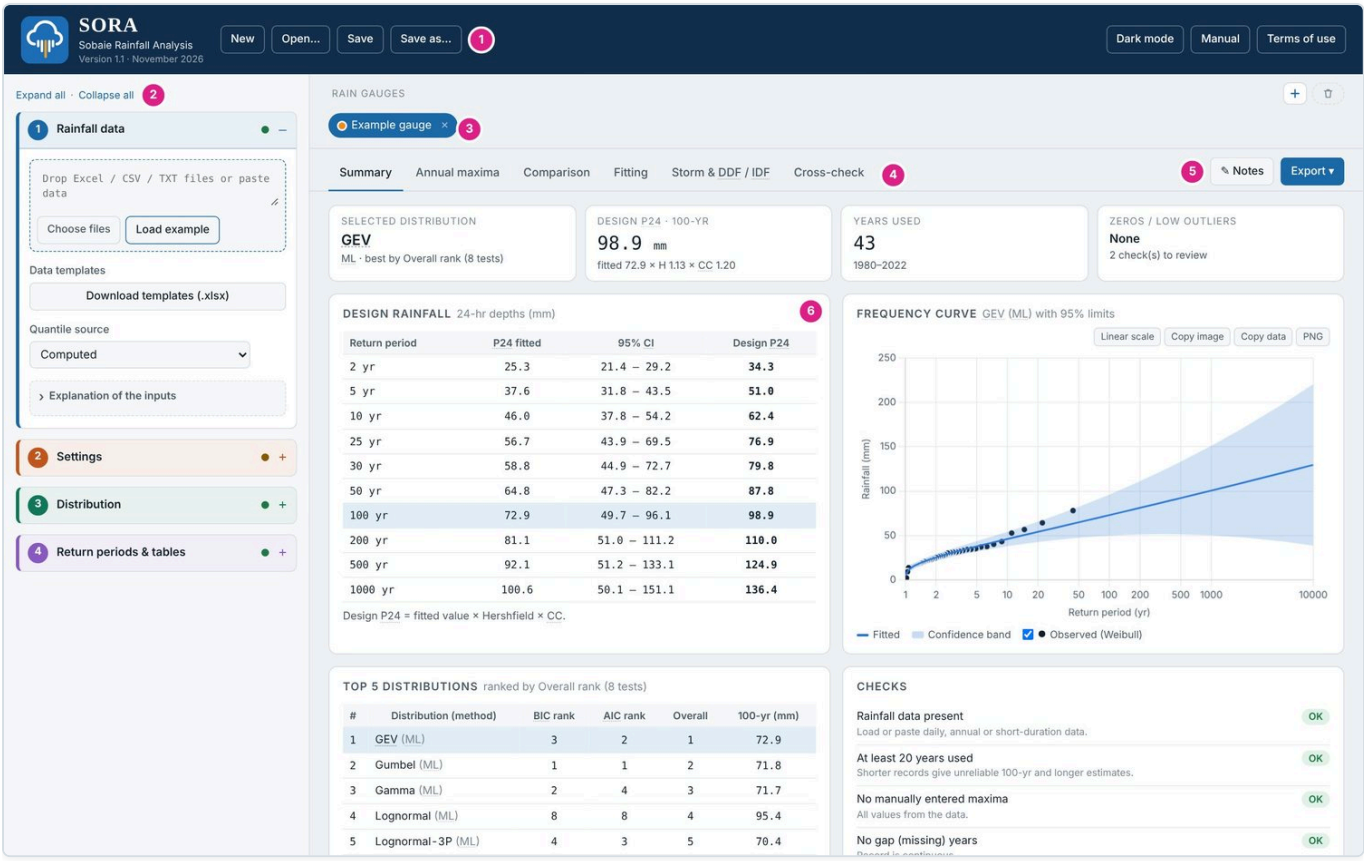


Figure 2.3 – The SORA screen with the example gauge

#	Element	What it does
1	Top bar	**New** (empty project; asks to save unsaved changes first), **Open...**, **Save**, **Save as...** project files (Chapter 8.2); **Dark mode** / **Light mode** (Chapter 9.2); **Assistant** (Chapter 9.3); **Manual** (this document, saved where you choose and opened); **Terms of use**.
2	Input steps	The four steps: **1 Rainfall data**, **2 Settings**, **3 Distribution**, **4 Return periods & tables**. Click a step heading to open or fold it (+ / −); *Expand all* · *Collapse all* open or fold every step. The coloured dot shows the state of the step: **green** OK, **amber** needs a look, **red** check, grey = no data. Drag the edge of the panel to make it wider (double-click resets).
3	Rain gauges	One chip per gauge with its status dot: click to open, double-click to rename, × to remove; + adds a gauge (Chapter 9).
4	Results tabs	Summary, Annual maxima, Comparison, Fitting, Storm & DDF / IDF, Cross-check (Chapter 7).
5	✎ Notes · Export ▾	Notes for this gauge (Chapter 9.1); the export panel for the PDF and Word reports and the Excel tables (Chapter 8.2).
6	Results	Key figures (selected distribution and method, design 24-hr depth at 100 yr, years used, zeros/low outliers and checks to review) and the tables and charts of the open tab.

Abbreviations – hover over any dotted abbreviation on the screen (ML, MoM, LM, AIC, BIC, AD, PPCC, GEV, LP3, CI, DDF, IDF, MGBT, EMA ...) to see what it stands for. **Undo** – **Ctrl+Z** undoes the last change (an input, a value, a removed gauge, an opened project) and **Ctrl+Y** redoes it; the last 40 steps are kept

while SORA is open. While you are typing in a box, Ctrl+Z first undoes the typing. **Frozen top** – while you scroll the results, the gauge chips, the result tabs and the buttons stay at the top of the window. Tool

Every change recalculates

There is no "Run" button. Changing any input recalculates the open gauge immediately; other gauges are recalculated in the background, shown by a progress indicator next to the gauges. On narrow windows (below 900 px) the steps are shown above the results.

3 Step 1 – Rainfall data

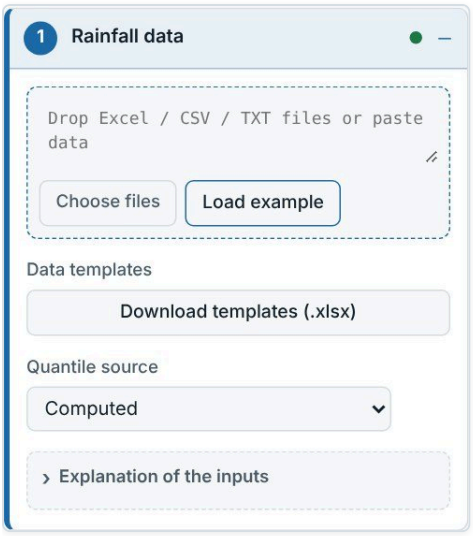


Figure 3.1 – Step 1

Upload or paste

Drop one or more Excel (.xlsx/.xls), CSV or TXT files (including rain-gauge report .txt files, Chapter 3.1) on the box, click **Choose files**, or paste cells copied from Excel and choose **Use for this gauge** or **Add as new gauge**. **Load example** loads a neutral demonstration record ("Example gauge", 43 years, 1980–2022) with demonstration settings – all eight tests ticked, every distribution fixed to ML (Log-Pearson III WRC, Generalized Pareto MoM – the methods of the software its imported quantiles come from) and sample imported quantiles on the Cross-check tab – so its results differ slightly from the default settings.

Data templates

Download templates (.xlsx) saves a workbook with a *Read me* sheet (the accepted layouts, their column order and the other header names SORA accepts) and one empty sheet per layout: *Daily table* (Year, Day, Jan ... Dec), *Monthly maxima* (Year, Jan ... Dec, Total), *Daily series* (Date, Rainfall), *Annual maxima* (Year, Max) and *Short-duration* (Date-time, Rainfall). Header names are labels only: SORA reads the column order. Fill a sheet (or copy it to a new file per gauge) and load it. Tool

Quantile source

Computed (default) designs with SORA's own quantiles. **Imported** designs with the quantiles pasted on the Cross-check tab (Chapter 7.6) – e.g. to reproduce a study made with other software.

Pasting from Excel

Select the cells in Excel (with or without the header and any empty columns on the left), press **Ctrl+C**, click the box and press **Ctrl+V**. Excel also places a picture of the cells on the clipboard; SORA ignores it and reads the text. Pasted data and dropped files give identical results.

Gauge naming

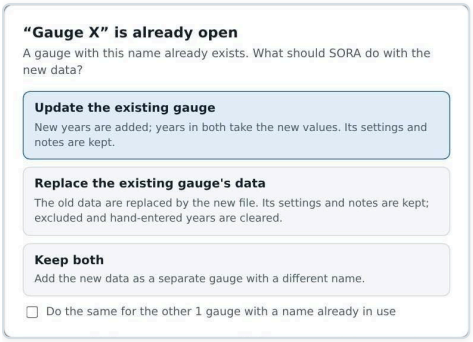


Figure 3.2 – A gauge name already in use

- A workbook with one data sheet → the gauge takes the **file name**.
- A workbook with several data sheets → one gauge per sheet, named after the **sheet**. Sheets without data (notes, charts) are ignored.
- **Name already in use** (a new file, sheet or project gauge with the name of an open gauge) → SORA asks what to do:
  - **Update the existing gauge** – new years are added and years in both take the new values (e.g. open 1980–2004 + new 2000–2024 → 1980–2024, with 2000–2004 from the new file). Settings and notes are kept; excluded or hand-entered marks on the replaced years are cleared. Daily, Year/Day and annual data of different layouts are combined as annual maxima; short-duration and daily data cannot be combined and stay as two gauges.
  - **Replace the existing gauge's data** – the new data replace the old; settings and notes are kept, excluded and hand-entered years are cleared (for a project file, the saved gauge replaces the open one with its own settings).
  - **Keep both** – the new gauge is added as "(2)", "(3)" … (also what Esc does).

  With several clashes, tick *Do the same for the other …* to answer once. Ctrl+Z undoes the result.

Double-click a gauge chip to rename the gauge. The layout of every table is detected automatically (Chapter 3.1); if a table is read wrongly, copy it into the matching template sheet. Tool

3.1 Accepted layouts (detected automatically)

Layout	Columns	How the annual maximum is obtained
Daily table	Year, Day, Jan … Dec (31 rows per year, or only the days with rain)	Largest daily value in each calendar year.
Monthly table	Year, Jan … Dec (one row per year, no Day column; a total column is ignored)	Largest of the 12 monthly values – use monthly **maximum daily** rainfall, not monthly totals. Step 1 shows a reminder.
Daily series	Date, rainfall	Largest daily value in each calendar year.
Annual maxima	Year, annual maximum	Used as given.
Short-duration (sub-daily)	Date-time, rainfall (e.g. hourly)	Largest **floating** 24-hour total in each year (a moving window of 24 hours measured on the time stamps, so gaps in the series are handled); completeness checked. Rows out of time order are sorted and repeated time stamps are read once.

Rain-gauge report (.txt)	“MONTHLY SUMMARY OF RAINS” text report: YEAR, DAY, JAN … DEC	Read as a daily table. Each value is placed in its month by its column position (a blank cell = no reading); values separated by tabs, commas or semicolons are read too. Titles, the STATION line, dashes, TOTALS lines and the closing code line are skipped. The gauge is named after the file.

Rules applied while reading Tool

- **Dates**: ISO (2020-03-15), day/month/year, Excel date numbers. A file is read as month/day/year (US) only when, in the first column of the first 5,000 rows, some second field is above 12 and no first field is (e.g. 12/25/2000).
- **Missing values**: blank cells, text (e.g. “trace”) and **negative values such as −999** are treated as missing, never as zero. When the data contain negative values, step 1 states how many and in which years, the years are marked negative on the Annual maxima tab, and the check *Negative values treated as missing* asks for review. For sub-daily data a missing time step does not count towards the year's completeness but adds 0 to the moving 24-hour total.
- **Missing years** are skipped, not treated as zero, and reported as gaps R21.
- At least **5 years** with rain above zero and **3 different values** are needed; otherwise SORA stops with a message. It also stops if fewer than 3 different values remain after zeros and low outliers are censored (then set *Zeros, low outliers & LP3 skew limit* Off in step 2).

3.2 Why “annual maximum series”

Design rainfall is estimated from the annual maximum series (AMS): one largest value per year, which gives an independent, identically distributed sample suitable for extreme-value analysis R5 R21 R18. Return period T relates to annual exceedance probability: $T = 1 / P(X > x)$.

3.3 Sub-daily data

For records finer than daily, the moving-window 24-hour maximum is close to the true 24-hour maximum, so the Hershfield factor defaults to **1.00** (Chapter 4). The time step is read from the dates. Years with any missing time step (completeness below 100 %) are marked amber as incomplete and listed for review. R10 R22

3.4 File size and reading time

SORA has no fixed row limit. An Excel sheet holds at most 1,048,576 rows (≈ 120 years of hourly data); CSV/TXT files are limited only by the computer memory (a few million rows). While files are loaded, a **loading bar** in the middle of the dimmed screen shows the overall percentage, the file being read (“File 3 of 100”) and the current stage (reading, opening the workbook, reading each sheet, finding the annual maxima). After loading, the analysis of the other gauges continues in the background (progress line in the gauge list). Typical times on a normal laptop: a daily table (500 rows) < 0.5 s; 46 years of hourly data (410,000 rows) ≈ 1–3 s, of which most is opening the Excel file. Saving large files as CSV reads faster. Tool

Files that could not be loaded

When some of the selected files (or sheets of a workbook) cannot be read, the others are still loaded and a **Not loaded** list appears above the results: the number of files not loaded out of those selected (e.g. “Not loaded: 4 of 6 files”), each file or sheet by name and the reason – *empty file*, *not a valid Excel file*, *password-protected*, *no rainfall table recognised* or *no sheet with a recognisable rainfall table*. The list stays until **Close**; **Copy list** copies it for Excel or e-mail. Fix the files and load them again – the gauges already loaded are kept. Tool

4 Step 2 – Settings

Step 2 holds three groups of settings – **Adjustment factors**, **Zeros & outliers** and **Design storm** – each with its own *Apply … to all gauges* button (shown when there are two or more gauges).

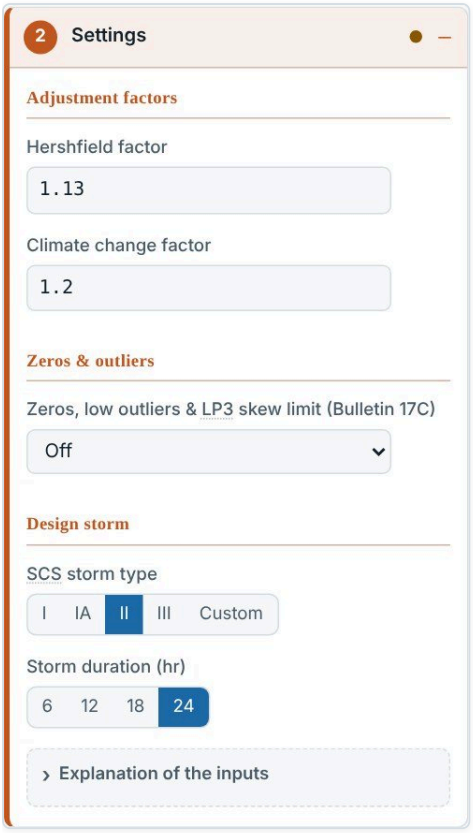


Figure 4.1 – Step 2

4.1 Adjustment factors

Hershfield factor (default 1.13 daily · 1.00 sub-daily)

Daily gauges are read at a fixed time (e.g. 09:00–09:00), so the largest fixed-day total is smaller than the true largest 24-hour total. Multiplying by 1.13 converts fixed-interval daily maxima to true 24-hour maxima R10 R22 R24. Use 1.00 if the project specifies no adjustment. The value is always editable.

Sub-daily data: the default is **1.00**, because the 24-hour total is taken from a moving window. Any other value shows an amber warning under the box and a REVIEW check, and the report states the value used. WMO-No. 1045 gives about **1.01** for a 24-hour total built from 24 hourly readings (1.02 for 5–8 readings, 1.03–1.04 for 2–4) R22 R24. The default switches automatically when data of the other type (daily ↔ sub-daily) are loaded into a gauge; a value you typed into a still-empty gauge is kept.

Climate change factor (default 1.20)

A uniform uplift on all design depths for future climate, as required by the project or the authority (e.g. 1.10–1.30). Heavy rainfall intensities are expected to rise by about 7 % per °C of warming (Clausius–Clapeyron scaling) R12. 1.00 = no allowance.

Basis of the factors

The report quotes a fixed basis next to each factor: *Hershfield (1961); WMO-No. 168 (2009)* for the Hershfield factor and *Project / authority requirement* for the climate change factor.

Apply factors to all gauges copies the climate change factor to every gauge and the Hershfield factor to the gauges with the same type of data (daily or short-duration). Tool

$$\text{Design P24(T)} = Q_T \times H \times CC \quad \text{where } Q_T = \text{best-estimate quantile of the selected distribution, H = Hershfield factor (default 1.00 for sub-daily), CC = climate-change factor} \tag{4.1}$$

Accepted input ranges

A value outside these ranges is refused with a message and the previous value is kept: Hershfield factor 1.00 – 2.00 · climate change factor 1.00 – 2.00. Tool

4.2 Zeros, low outliers and the LP3 skew limit

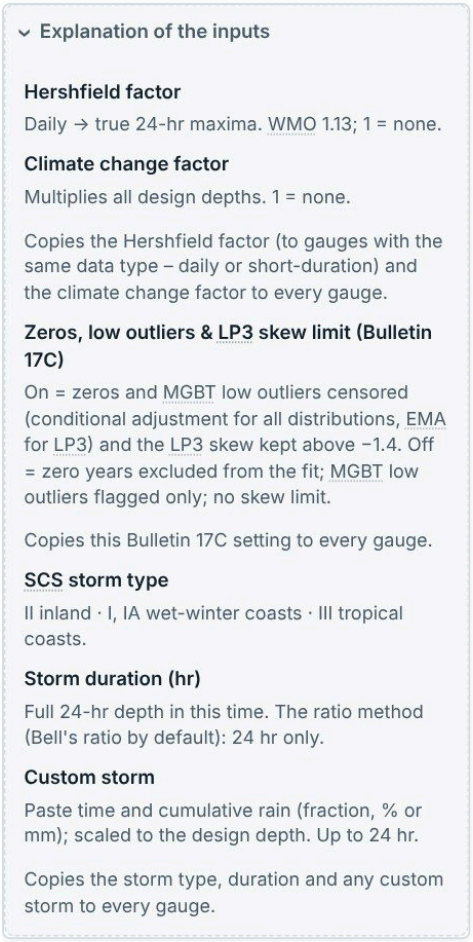


Figure 4.2 – Every step ends with an "Explanation of the inputs" section (step 2 shown)

High outliers – Z-score

Each year gets Z = (P − mean) / SD; years with |Z| above **3** (fixed) are flagged for review on the Annual maxima tab. They are **not** removed automatically – a real extreme storm is the most important information in the record R21 R6.

Low outliers – Multiple Grubbs–Beck test (MGBT)

Potentially influential low floods (PILFs) can distort the fit of the upper tail. MGBT tests the k smallest values on log scale: an outward sweep (α = 0.5 %) from the median and an inward sweep (α = 10 %) from the smallest value; the larger k is used R4 R6. With the option **On** these years are censored (treated as "below threshold"), not deleted.

One option: Zeros, low outliers & LP3 skew limit (Bulletin 17C)

A single On/Off setting (default **Off**) switches the Bulletin 17C treatment of zeros and low outliers and the LP3 skew lower limit together. The LP3 skew is always the station skew (no regional skew). **Apply zeros & outliers to all gauges** copies this setting to every gauge. Tool

Setting	Treatment	Reference
On	MGBT identifies the low outliers; zero years and low outliers are **censored together** (known only to be below the threshold). All distributions are fitted to the remaining years and combined with the probability of a censored year p_o (conditional adjustment, Eq. 4.2). Log-Pearson III uses the **Expected Moments Algorithm** (EMA) of Bulletin 17C, and its skew is kept above −1.4 (Chapter 4.3).	R6 R4 R14 R7
Off (default)	Zero years are **excluded** and the distributions are fitted to the years with rain. MGBT still runs, but its low outliers are only **flagged** (check *Low outliers (MGBT, 17C)*), not censored. No skew limit. The Summary shows how many zero years were excluded from the fit.	Tool

$$P(X \le x) = p_0 + (1 - p_0) \cdot F(x) \quad \Rightarrow \quad x_T = F^{-1}[\,(1 - 1/T - p_0) / (1 - p_0)\,], \quad x_T = 0 \text{ if } 1 - 1/T \le p_0 \quad (p_0 = \text{censored years / all years}) \tag{4.2}$$

> **Note**
>
> Inside MGBT a zero is represented by a tiny value (as in the USGS code), so it is always identified as a low outlier and censored. Records with many zeros can have up to half of the years censored; with Off, a record with many dry years is fitted to its wet years only, which overstates the frequent (short return period) depths. R4

4.3 Log-Pearson III, EMA and skew

- **EMA** (Expected Moments Algorithm) estimates the LP3 moments when some years are only known to be below a threshold (censored interval 0 – T). It is the Bulletin 17C method R3 R6; SORA iterates it with SQUAREM acceleration and continues with plain iterations until convergence R19. If EMA does not converge, the LP3 fit status reads *EMA not converged – check* and, if LP3 is the design distribution, the status is flagged CHECK.
- **Skew**: the station skew is used (no regional skew weighting). A skew of almost exactly zero ($|G| < 0.0001$) is set to ±0.0001, so LP3 gives the lognormal values instead of failing. Tool
- **Skew lower limit** (with the option On): keeps the LP3 skew above −1.4 in every EMA iteration, as required by Bulletin 17C (England et al. 2019, Appendix 7, p. 81) R6. Off removes the limit.

4.4 Design storm

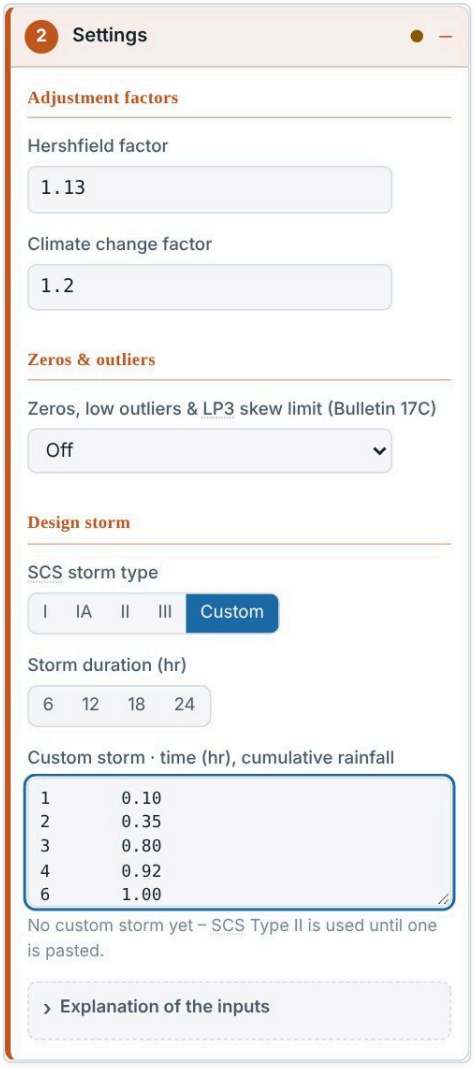


Figure 4.3 – Step 2 with a custom storm pasted

SCS (NRCS) storm types

The design 24-hour depth is distributed in time with the NRCS dimensionless cumulative rainfall curves at a 0.1-hour step R16 R17:

- **Type I, IA** – Pacific maritime climates with wet winters and dry summers.
- **Type II** – most inland regions; the most intense curve (peak at 12 h).
- **Type III** – coastal areas affected by tropical storms.

Storm duration (hr)

6, 12, 18 or 24 h. The full design 24-hour depth is placed within the chosen duration using the most intense part of the 24-hour curve, rescaled to end at 1.0. The ratio method (Bell's ratio by default) is calculated for the 24-hour storm only. Tool

Custom storm

Choose **Custom** as the storm type and paste time (h) and cumulative rainfall (fraction, % or mm) of any local or national design pattern (up to 24 h). Columns may be separated by a tab (pasted from Excel), a semicolon, spaces or a single comma (*2,40*); decimal commas are read when the columns are separated otherwise (*0,5 0,2*). SORA sorts the points, interpolates to 0.1 h and scales the curve to end at 1.0, then applies it to every design depth; the line under the box confirms what was read. Tool

Return period shown in charts

Always **100 yr** – used for the key figures, hyetograph, mass curve and comparison charts. If 100 yr is removed from the return periods, the nearest one is used.

Defaults: SCS Type II, 24-hour storm. **Apply design storm to all gauges** copies the storm type, the duration and any custom storm to every gauge. Tool

$$\text{Cumulative depth } P(t) = \text{Design P24} \times C(t), \quad \text{increment } \Delta P_i = \text{Design P24} \times [C(t_i) - C(t_{i-1})], \quad \Sigma\,\Delta P = \text{Design P24} \quad \text{(checked)} \tag{4.3}$$

5 Step 3 – Distribution

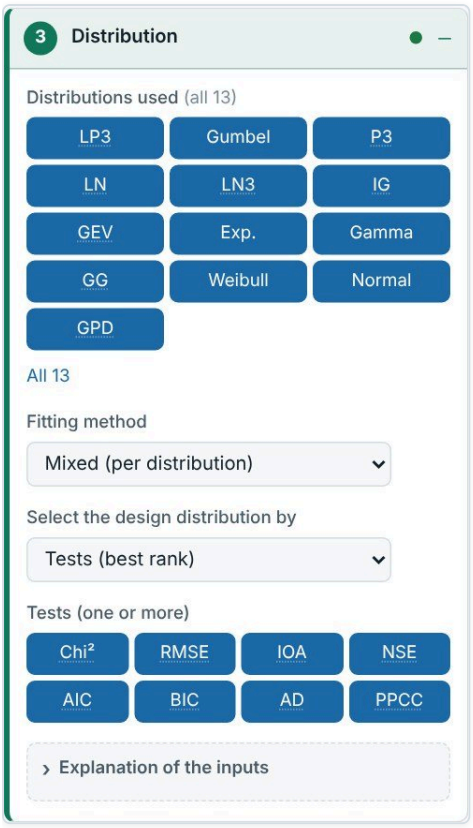


Figure 5.1 – Step 3

Thirteen probability distributions are fitted to the annual maxima (Appendix A). They are listed in every table in this order: Log-Pearson III, Gumbel (EV1), Pearson III, Lognormal, Lognormal-3P, Inverse Gamma, GEV, Exponential, Gamma, Generalized Gamma, Weibull, Normal and Generalized Pareto R18 R11 R21.

Distributions used

All 13 are on by default. Click a distribution to switch it off: it is then not fitted, ranked, selected or reported (screen, Word/PDF report and Excel) until it is switched on again; *All 13* switches every one back on. At least two stay on. With few distributions the Quantiles and Ranking tables of the Fitting tab sit side by side, and the Comparison tab lists only the distributions used (e.g. a top 4). If the manual choice is switched off, the best by BIC is used and the check *Distributions used* asks for review.

Fitting method

Sets the parameter-estimation method of all distributions at once: **Auto (best)** (default – the method that ranks best on the selected tests), **MoM**, **ML** or **LM** (Chapter 5.4). A distribution without the chosen method uses Auto; Log-Pearson III always uses WRC (method of moments of the logarithms, or EMA). Methods can also be set for single distributions on the Fitting tab; the list then shows *Mixed (per distribution)*.

Select the design distribution by

- **Tests (best rank)** – tick one or more of the eight tests under *Ranking*; the distribution with the lowest **sum of ranks** over the ticked tests is selected. The default selection is the first six tests (= overall rank). Shortcuts: *All* (all eight tests), *GOF* (Chi², RMSE, IOA, NSE), *AIC + BIC*, *AD + PPCC*.
- **Manual choice** – pick the distribution yourself (e.g. one required by a national guideline).

Apply step 3 to all gauges (two or more gauges) copies the distributions used, the ticked tests – e.g. only AIC + BIC – or the manual distribution, the quantile source (step 1) and the methods set on the Fitting tab to every gauge.

5.1 Plotting position

Weibull: $P(\text{exceedance}) = m / (N + 1)$, $T = (N + 1) / m$ (m = rank in descending order) (5.1)

The Weibull formula is unbiased for exceedance probabilities of any distribution and is the usual choice in hydrology R25 R5 R7.

5.2 The eight selection tests

Test	Formula (O = observed, E = distribution value at the same T)	Better	Reference
Chi-squared	$\Sigma\ (O - E)^2 / E$	lower	R18
RMSE	$\sqrt{[\ \Sigma\ (O - E)^2 / n\ ]}$	lower	R18
IOA (Willmott)	$1 - \Sigma(O - E)^2 / \Sigma(\lvert E - \bar{O}\rvert + \lvert O - \bar{O}\rvert)^2$	higher	R27
NSE	$1 - \Sigma(O - E)^2 / \Sigma(O - \bar{O})^2$	higher	R15
AIC	$-2 \ln L + 2k$	lower	R1
BIC	$-2 \ln L + k \ln n$	lower	R20
Anderson–Darling (AD)	$A^2 = -n - (1/n)\ \Sigma\ (2i - 1)[\ln F(x_i) + \ln(1 - F(x_{n+1-i}))]$, x ascending	lower	R29 R30
PPCC	correlation r between the ordered O and E	higher	R31

L = likelihood of all fitted years, k = number of parameters, n = sample size. AIC and BIC penalise extra parameters, so a 3-parameter distribution must fit clearly better to rank above a 2-parameter one. **AD** uses the fitted CDF F at every fitted year (zeros and censored low outliers excluded; for LP3 with EMA, F is conditional on exceeding the censoring threshold) and gives extra weight to the tails, where the design values come from. All eight tests use every fitted year (all observations). AD and PPCC are ranking criteria only; SORA does not use them to reject distributions. The overall rank is always the sum of the ranks of the first six tests, whichever tests select the distribution.

5.3 Ranking and selection Tool

1. Each distribution is ranked 1–13 on each test (ties share the better rank; a test that cannot be computed counts as the worst rank, 13).
2. The ranks of the selected tests are added: **score**. The lowest score wins; equal scores are resolved in this fixed order of the distributions (Weibull, Gamma, Gumbel, Pearson III, Lognormal-3P, GEV, Normal, Exponential, Lognormal, Log-Pearson III, Generalized Pareto, Generalized Gamma, Inverse Gamma).
3. The **Top 5** table and curve chart are ordered by this score. With the default six tests this is the overall rank.

5.4 Estimation methods

Method	Principle	Reference
ML	Maximum likelihood – parameters that make the observed sample most probable; solved numerically. Efficient for larger samples.	R18 R13
ML – Gumbel	For Gumbel, maximum likelihood is bias-corrected for short records (Fiorentino & Gabriele 1984): $\alpha^* = \hat{\alpha} \cdot n/(n - 0.8)$; u* from the ML location equation at α*, minus $0.7 \cdot \alpha^*/n$. Standard errors by the delta method on these estimators.	R32 R33

MoM	Method of moments – fitted mean, standard deviation (n − 1) and, for 3-parameter distributions, skewness equal the sample values.	R13 R5
LM	L-moments – linear combinations of ordered data; robust to outliers and preferred for short records. GEV shape from Hosking et al. (1985) refined by Newton iteration.	R8 R9 R11
WRC	Log-Pearson III by moments of the base-10 logarithms (Bulletin 17), or EMA when years are censored (Chapter 4.3).	R6 R26

Which method is used?

Default: "Auto (best)" – every available method is fitted and each distribution uses the method whose fit ranks best on the selected tests (ranks pooled over all distribution–method pairs). The *Fitting method* list in step 3 – or the list in the heading of the Method column on the Fitting tab – sets all distributions at once, and any method can be fixed per distribution on the Fitting tab – a distribution without that method is set to Auto and a message names it. Log-Pearson III (WRC / EMA) and Generalized Gamma (ML) have a single method: their list holds only that method (no Auto). Tool

6 Step 4 – Return periods & tables

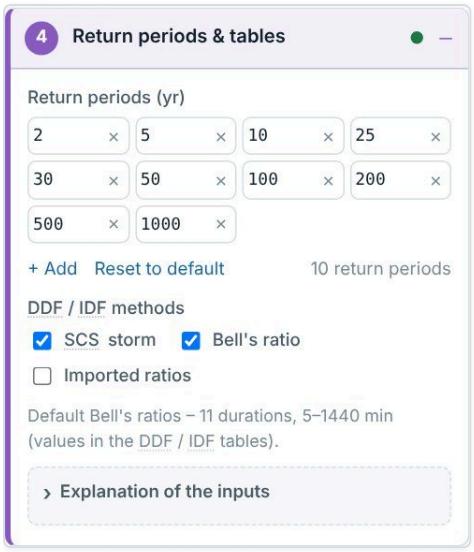


Figure 6.1 – Step 4

Input	Meaning
Return periods (yr)	Default 2, 5, 10, 25, 30, 50, 100, 200, 500 and 1000 yr. **+ Add**, × to remove, **Reset to default**; sorted automatically; 2 – 24 values, each above 1 and up to 10,000 yr; duplicates refused.
DDF durations	Fixed: 5, 10, 15, 20, 30, 60, 120, 180, 360, 720 and 1440 min (DDF/IDF tables).
DDF / IDF methods	Tick the **SCS storm** method, the ratio method (**Bell's ratio**) or both (default: both). A method that is unticked is left out of the Storm & DDF / IDF tab, the Word/PDF report and the Excel tables; the IDF comparison chart appears only when both are shown.
Imported ratios	Leave unticked for Bell's built-in ratios (listed in the DDF / IDF tables). Tick it to use your own table: type the **Name of the ratio method** and paste two columns copied from Excel into **Ratio table** (duration · ratio Pt/P24). Durations may be in minutes or hours (read as hours when the header says hr or the last duration is 24), ratios as fractions or %; the table must reach 24 hr (1440 min) with ratio 1.0 and the ratios must not decrease. It is applied automatically. The name you type replaces “Bell's ratio” everywhere – tabs, checks, charts, Word/PDF report and Excel tables – and the Bell (1969) reference is then left out of the report. Untick to return to Bell's ratios.
Apply step 4 to all gauges	Copies the return periods, the DDF / IDF methods and the ratio table with its name to every gauge.

Project name, prepared-by and report-cover details are not part of the steps: the gauge name is used as the title, and the document-control fields are typed in the export panel (Chapter 8.2).

Bell's ratios

Bell (1969) showed that short-duration depths are nearly constant fractions of a reference depth across regions R2. The default Pt/P24 values are SORA's built-in set; replace them (and rename the method) with ratios for your region where available. Tool

7 Results tabs

Copying tables and charts

Click a **column header** to select that column; **Shift-click** another header to extend the selection to a block of columns. Click a **table title** to select the whole table. Selected cells are highlighted; press **Ctrl+C** (⌘C on Mac) and paste in Excel – values arrive tab-separated with their headers (ticked years copy as Yes/No). Click elsewhere or press Esc to clear.

Chart buttons (above every chart): *Linear scale / Log scale* switches logarithmic axes to linear and back (linear return periods are shown up to 100 yr, durations up to 1,440 min, so the frequent range stays readable); *Copy image* copies the chart as a picture for Word, PowerPoint or Excel, with a title (gauge and chart name) above and a legend of the series shown below (where copying is blocked, a PNG file is saved instead); *Copy data* copies the values of the curves and points shown at that moment – unticked curves and hidden observed points are left out with their headers – paste in Excel and use *Insert → Chart* for a fully editable chart; *PNG* saves the same picture (title and legend included) as a file.

Chart values on hover: the list shown when hovering over a chart gives the value of every curve shown at that return period, duration or year, ordered from the highest to the lowest value. On the frequency and Comparison curves, hovering an observed point shows its value, year and plotting-position return period.

Row and column highlight: click any cell of any table to highlight its row and column (the cell is outlined); click the same cell again, click outside the table or press Esc to remove it. Tool

7.1 Summary

Key figures – selected distribution and method, design P24 at the chart return period with its factors, years used, zeros/low outliers and the number of checks to review.

Design 24-hr rainfall – fitted P24, its 95 % confidence interval and design P24 (Eq. 4.1) for every return period.

Frequency curve – selected distribution on a log return-period axis, shaded confidence band, observed annual maxima at their Weibull return periods (Eq. 5.1). The *Observed* tick under the chart shows or hides the observed points (also on the Comparison chart).

Below: the top 5 table with BIC, AIC and overall ranks, the distribution curves with the picker (7.4) and the full list of checks (Chapter 8.1).

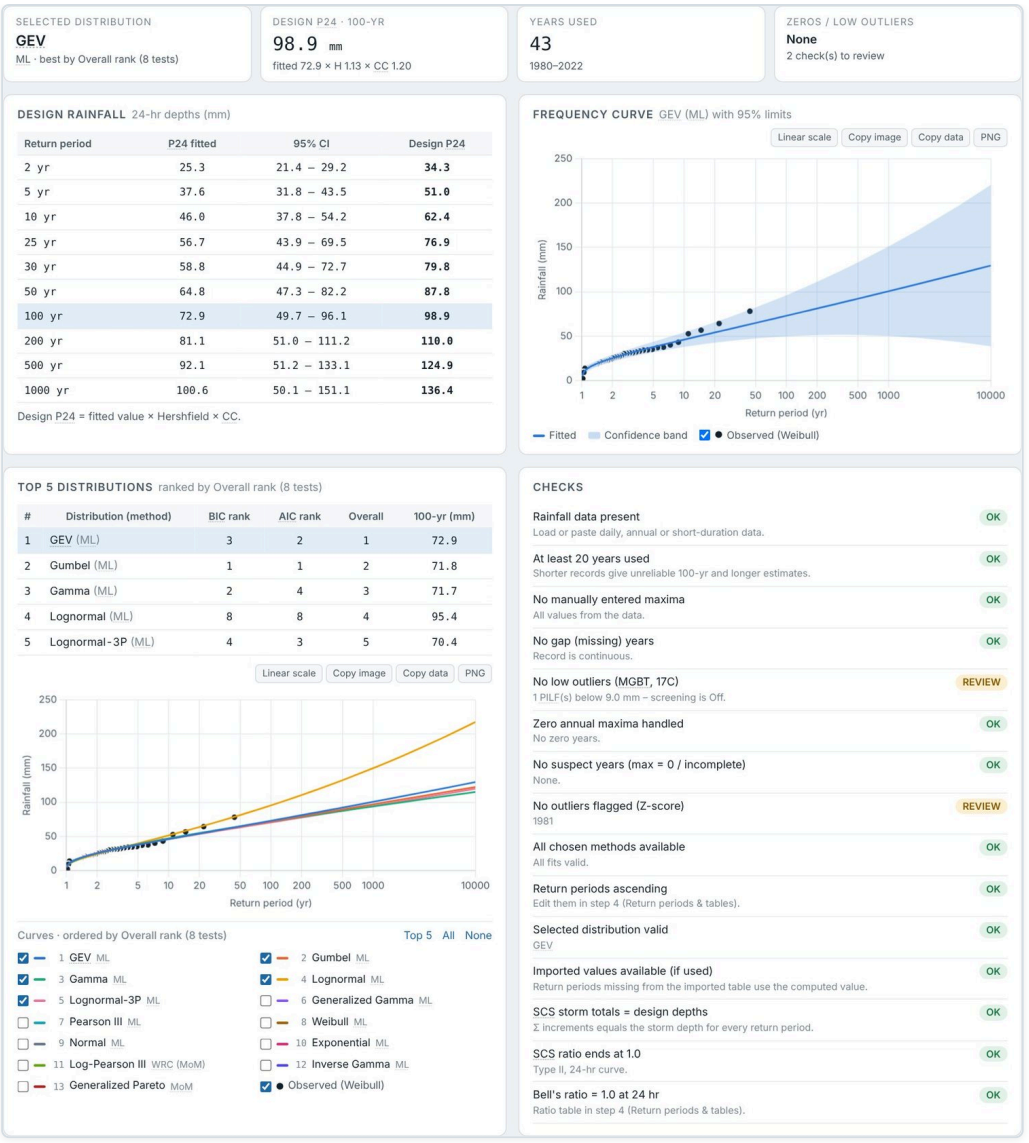


Figure 7.1 – Summary tab

7.2 Annual maxima

Annual maximum series – one row per year; untick a year to exclude it (the row is struck through). The *Check* column shows OK, Outlier, Low outlier, Max = 0, Incomplete, Manual or Missing, next to the Z-score. The tab has two equal columns: the year table on the left (it scrolls when there are many years and ends level with the Statistics table) and the statistics and calendar on the right.

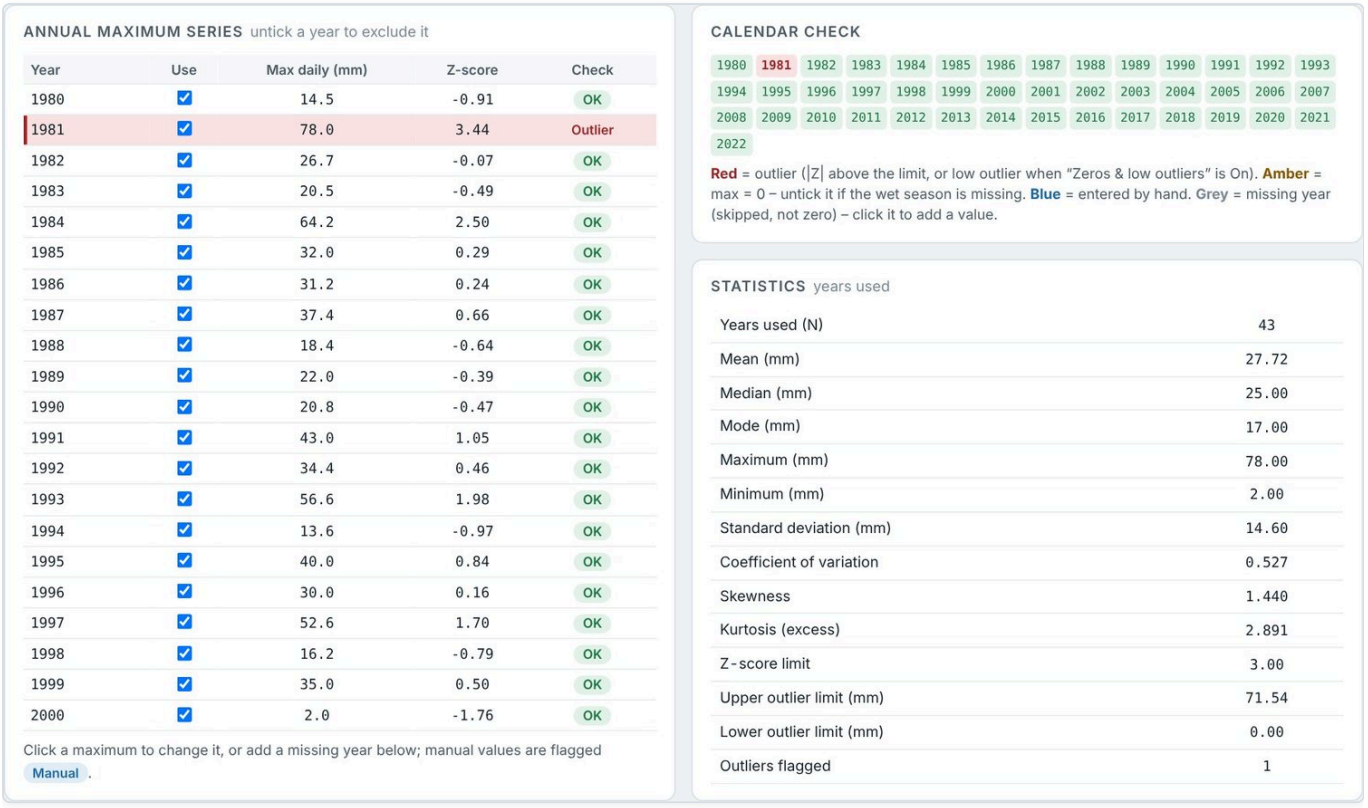
Calendar check – one chip per year from the first to the last year.

Colour code – the table rows and the calendar chips use the same colours; rows that are OK keep the normal colour:

Colour	Meaning
red	Outlier: |Z| above 3, or low outlier when *Zeros & low outliers* is On – verify the value.
amber	Maximum = 0, or sub-daily year with missing time steps – check the wet season.
blue	Entered by hand (*Manual*).
grey	Missing year – skipped (not zero) until a value is entered.

Editing – click a maximum, type a value and press Enter (Esc cancels; ↺ or an empty box restores the data value). To add a missing year, click its grey chip or type the year in the last row, enter the maximum and click **Add year**; years outside the record can be added the same way. Manual values are listed in the checks, the report and the Excel tables with the original value. Tool

Statistics – N, mean, median, mode, max, min, SD, coefficient of variation, skewness, excess kurtosis and outlier limits.



ANNUAL MAXIMUM SERIES untick a year to exclude it

Year	Use	Max daily (mm)	Z-score	Check
1980	☑	14.5	-0.91	OK
1981	☑	78.0	3.44	Outlier
1982	☑	26.7	-0.07	OK
1983	☑	20.5	-0.49	OK
1984	☑	64.2	2.50	OK
1985	☑	32.0	0.29	OK
1986	☑	31.2	0.24	OK
1987	☑	37.4	0.66	OK
1988	☑	18.4	-0.64	OK
1989	☑	22.0	-0.39	OK
1990	☑	20.8	-0.47	OK
1991	☑	43.0	1.05	OK
1992	☑	34.4	0.46	OK
1993	☑	56.6	1.98	OK
1994	☑	13.6	-0.97	OK
1995	☑	40.0	0.84	OK
1996	☑	30.0	0.16	OK
1997	☑	52.6	1.70	OK
1998	☑	16.2	-0.79	OK
1999	☑	35.0	0.50	OK
2000	☑	2.0	-1.76	OK

Click a maximum to change it, or add a missing year below; manual values are flagged Manual.

CALENDAR CHECK

1980 1981 1982 1983 1984 1985 1986 1987 1988 1989 1990 1991 1992 1993
1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007
2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021
2022

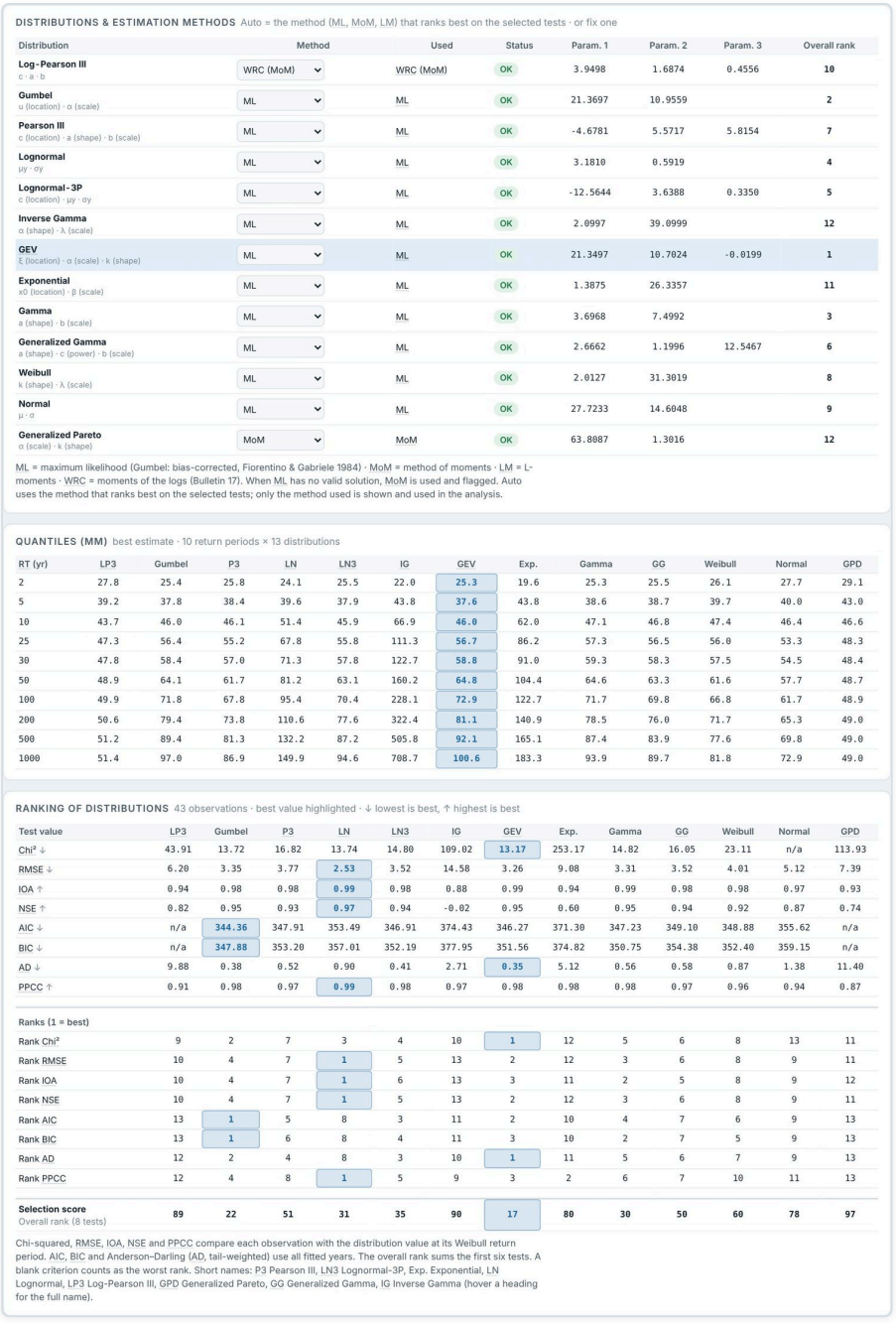
Red = outlier (|Z| above the limit, or low outlier when "Zeros & low outliers" is On). Amber = max = 0 – untick it if the wet season is missing. Blue = entered by hand. Grey = missing year (skipped, not zero) – click it to add a value.

STATISTICS years used

Years used (N)	43
Mean (mm)	27.72
Median (mm)	25.00
Mode (mm)	17.00
Maximum (mm)	78.00
Minimum (mm)	2.00
Standard deviation (mm)	14.60
Coefficient of variation	0.527
Skewness	1.440
Kurtosis (excess)	2.891
Z-score limit	3.00
Upper outlier limit (mm)	71.54
Lower outlier limit (mm)	0.00
Outliers flagged	1

Figure 7.2 – Annual maxima of the example gauge; 1981 is a high outlier (red)

7.3 Fitting



DISTRIBUTIONS & ESTIMATION METHODS — Auto = the method (ML, MoM, LM) that ranks best on the selected tests · or fix one

Distribution	Method	Used	Status	Param. 1	Param. 2	Param. 3	Overall rank
Log-Pearson III c · a · b	WRC (MoM)	WRC (MoM)	OK	3.9498	1.6874	0.4556	10
Gumbel u (location) · α (scale)	ML	ML	OK	21.3697	10.9559		2
Pearson III c (location) · a (shape) · b (scale)	ML	ML	OK	-4.6781	5.5717	5.8154	7
Lognormal μy · σy	ML	ML	OK	3.1810	0.5919		4
Lognormal-3P c (location) · μy · σy	ML	ML	OK	-12.5644	3.6388	0.3350	5
Inverse Gamma α (shape) · λ (scale)	ML	ML	OK	2.0997	39.0999		12
GEV ξ (location) · α (scale) · k (shape)	ML	ML	OK	21.3497	10.7024	-0.0199	1
Exponential x0 (location) · β (scale)	ML	ML	OK	1.3875	26.3357		11
Gamma a (shape) · b (scale)	ML	ML	OK	3.6968	7.4992		3
Generalized Gamma a (shape) · c (power) · b (scale)	ML	ML	OK	2.6662	1.1996	12.5467	6
Weibull k (shape) · λ (scale)	ML	ML	OK	2.0127	31.3019		8
Normal μ · σ	ML	ML	OK	27.7233	14.6048		9
Generalized Pareto α (scale) · k (shape)	MoM	MoM	OK	63.8087	1.3016		12

ML = maximum likelihood (Gumbel: bias-corrected, Fiorentino & Gabriele 1984) · MoM = method of moments · LM = L-moments · WRC = moments of the logs (Bulletin 17). When ML has no valid solution, MoM is used and flagged. Auto uses the method that ranks best on the selected tests; only the method used is shown and used in the analysis.

QUANTILES (MM) — best estimate · 10 return periods × 13 distributions

RT (yr)	LP3	Gumbel	P3	LN	LN3	IG	GEV	Exp.	Gamma	GG	Weibull	Normal	GPD
2	27.8	25.4	25.8	24.1	25.5	22.0	25.3	19.6	25.3	25.5	26.1	27.7	29.1
5	39.2	37.8	38.4	39.6	37.9	43.8	37.6	43.8	38.6	38.7	39.7	40.0	43.0
10	43.7	46.0	46.1	51.4	45.9	66.9	46.0	62.0	47.1	46.8	47.4	46.4	46.6
25	47.3	56.4	55.2	67.8	55.8	111.3	56.7	86.2	57.3	56.5	56.0	53.3	48.3
30	47.8	58.4	57.0	71.3	57.8	122.7	58.8	91.0	59.3	58.3	57.5	54.5	48.4
50	48.9	64.1	61.7	81.2	63.1	160.2	64.8	104.4	64.6	63.3	61.6	57.7	48.7
100	49.9	71.8	67.8	95.4	70.4	228.1	72.9	122.7	71.7	69.8	66.8	61.7	48.9
200	50.6	79.4	73.8	110.6	77.6	322.4	81.1	140.9	78.5	76.0	71.7	65.3	49.0
500	51.2	89.4	81.3	132.2	87.2	505.8	92.1	165.1	87.4	83.9	77.6	69.8	49.0
1000	51.4	97.0	86.9	149.9	94.6	708.7	100.6	183.3	93.9	89.7	81.8	72.9	49.0

RANKING OF DISTRIBUTIONS — 43 observations · best value highlighted · ↓ lowest is best, ↑ highest is best

Test value	LP3	Gumbel	P3	LN	LN3	IG	GEV	Exp.	Gamma	GG	Weibull	Normal	GPD
Chi² ↓	43.91	13.72	16.82	13.74	14.80	109.02	13.17	253.17	14.82	16.05	23.11	n/a	113.93
RMSE ↓	6.20	3.35	3.77	2.53	3.52	14.58	3.26	9.08	3.31	3.52	4.01	5.12	7.39
IOA ↑	0.94	0.98	0.98	0.99	0.98	0.88	0.99	0.94	0.99	0.98	0.98	0.97	0.93
NSE ↑	0.82	0.95	0.93	0.97	0.94	-0.02	0.95	0.60	0.95	0.94	0.92	0.87	0.74
AIC ↓	n/a	344.36	347.91	353.49	346.91	374.43	346.27	371.30	347.23	349.10	348.88	355.62	n/a
BIC ↓	n/a	347.88	353.20	357.01	352.19	377.95	351.56	374.82	350.75	354.38	352.40	359.15	n/a
AD ↓	9.88	0.38	0.52	0.90	0.41	2.71	0.35	5.12	0.56	0.58	0.87	1.38	11.40
PPCC ↑	0.91	0.98	0.97	0.99	0.98	0.97	0.98	0.98	0.98	0.97	0.96	0.94	0.87

Ranks (1 = best)													
Rank Chi²	9	2	7	3	4	10	1	12	5	6	8	13	11
Rank RMSE	10	4	7	1	5	13	2	12	3	6	8	9	11
Rank IOA	10	4	7	1	6	13	3	11	2	5	8	9	12
Rank NSE	10	4	7	1	5	13	2	12	3	6	8	9	11
Rank AIC	13	1	5	8	3	11	2	10	4	7	6	9	13
Rank BIC	13	1	6	8	4	11	3	10	2	7	5	9	13
Rank AD	12	2	4	8	3	10	1	11	5	6	7	9	13
Rank PPCC	12	4	8	1	5	9	3	2	6	7	10	11	13
Selection score Overall rank (8 tests)	**89**	**22**	**51**	**31**	**35**	**90**	**17**	**80**	**30**	**50**	**60**	**78**	**97**

Chi-squared, RMSE, IOA, NSE and PPCC compare each observation with the distribution value at its Weibull return period. AIC, BIC and Anderson–Darling (AD, tail-weighted) use all fitted years. The overall rank sums the first six tests. A blank criterion counts as the worst rank. Short names: P3 Pearson III, LN3 Lognormal-3P, Exp. Exponential, LN Lognormal, LP3 Log-Pearson III, GPD Generalized Pareto, GG Generalized Gamma, IG Inverse Gamma (hover a heading for the full name).

Figure 7.3 – Fitting tab: methods, parameters, quantiles and the ranking of distributions

- **Distributions & estimation methods** – method per distribution (Auto or fixed; the list in the column heading sets all at once), method actually used (only the method used is shown and used; the other methods tried by Auto are not reported), status (e.g. “ML not valid → MoM used”), parameters and overall rank.
- **Quantiles** – best estimate for every return period and distribution (the 95 % limits are shown on the Summary and in the report).
- **Ranking of distributions** – the eight test values (↓ lowest is best, ↑ highest is best), then (after a gap) their ranks under *Ranks (1 = best)* and the selection score. Test values are shown with two decimals; ranks use the full precision. The best value and every rank 1 are shown in bold blue in a shaded box; the selected distribution is shaded the same way in the quantile table. Column headings use short names (P3, LN3, Exp., LN, LP3, GPD, GG, IG; hover for the full name); parameter names are shown under each distribution name.

Confidence limits

Confidence limits are always shown at the 95 % level (z = 1.96); design values use the best estimate.

$$x_T \pm 1.96 \cdot SE(x_T), \quad SE^2 = \nabla x_T^{T} \cdot I^{-1} \cdot \nabla x_T \quad \text{(delta method, observed information I)} \tag{7.1}$$

- ML fits: delta method with the numerically computed observed Fisher information R18 R13.
- Normal: SE = s·√[(1 + z²/2)/n]; Lognormal: $x_T \cdot s_y \cdot \sqrt{[(1 + z^2/2)/n]}$; Gumbel (moments): SE = (s/√n)·√(1 + 1.1396K + 1.1K²); Pearson III and LP3 (moments): Kite's frequency-factor formula R13; Exponential: closed form; Generalized Pareto by moments: delta method on the sample mean and standard deviation.
- GEV and Generalized Pareto by ML with shape k > 0.5: maximum likelihood is non-regular and no valid limits exist – shown as n/a R23.
- Limits are also shown as n/a when the delta method has no usable derivative (fit at a parameter bound) or gives a standard error larger than three times the quantile – such limits are meaningless (unstable fit). Tool

7.4 Comparison (top 5 distributions) and the curve picker

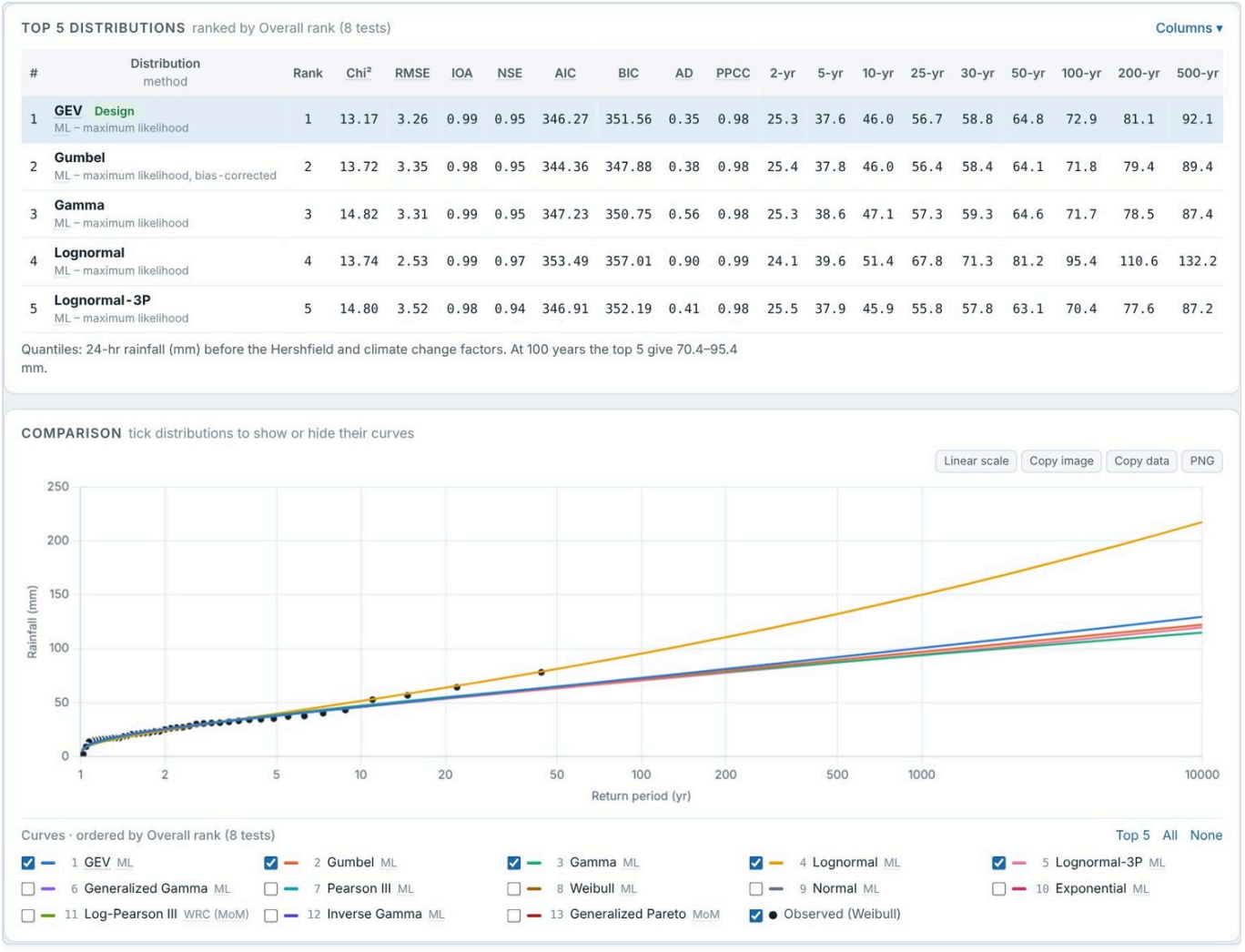


Figure 7.4 – Comparison tab: top 5 table, comparison chart and curve picker

The **Comparison** tab lists the five best distributions by the selected tests (fewer when fewer distributions are used in step 3) and compares their curves – the place to prepare the comparison tables and charts for a report. Under each name is the estimation method used for its parameters (e.g. *ML – maximum likelihood*). If the design distribution was chosen manually and is not in the top 5 it is shown below them.

Drag a column header to move that column. **Columns ▾** chooses what the table shows: rank, score (sum of ranks), overall rank, fit status, parameters, the 95 % interval at 100 yr, the values (2 decimals) or ranks of any of the eight tests, and any return periods. The **default** is: distribution and method, rank, the values of the selected tests and all return periods. *Reset to default* returns this gauge to the default; *Use as default for all gauges* makes the current columns the default for every gauge; *Restore the original default* brings the built-in columns back. Quantiles are 24-hour rainfall before the Hershfield and climate change factors. Tool

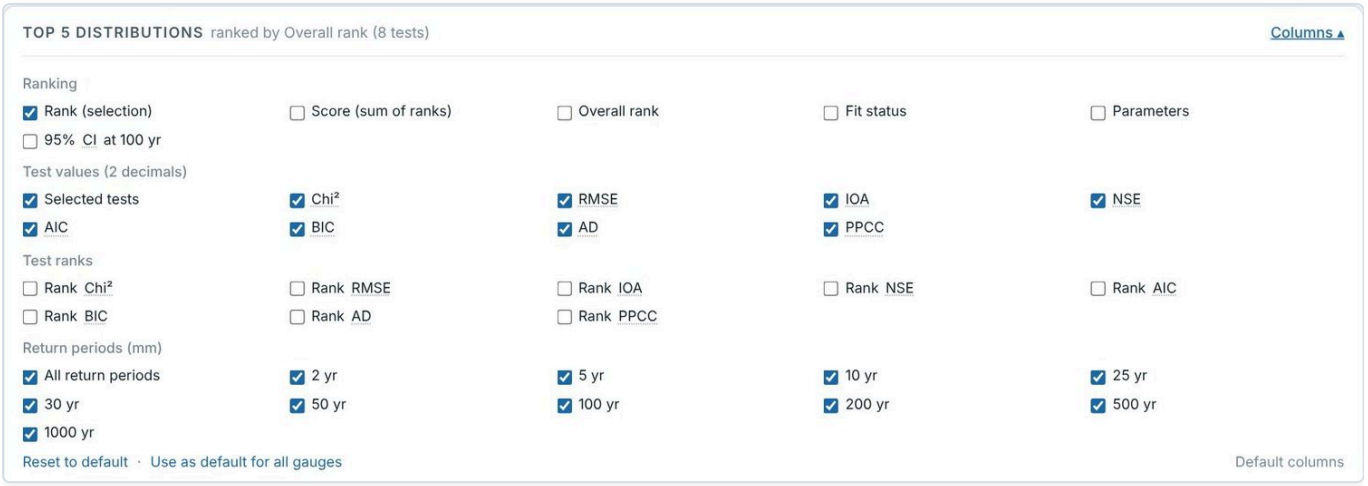


Figure 7.5 – Columns ▾: choosing the columns of the Top 5 table

Curve picker – under the comparison chart all distributions used are listed in rank order with their colour and method. The first five are ticked; tick or untick any distribution, or use *Top 5 · All · None*. A wide spread between well-ranked curves at long return periods means the design value is sensitive to the choice of distribution.

7.5 Storm & DDF / IDF

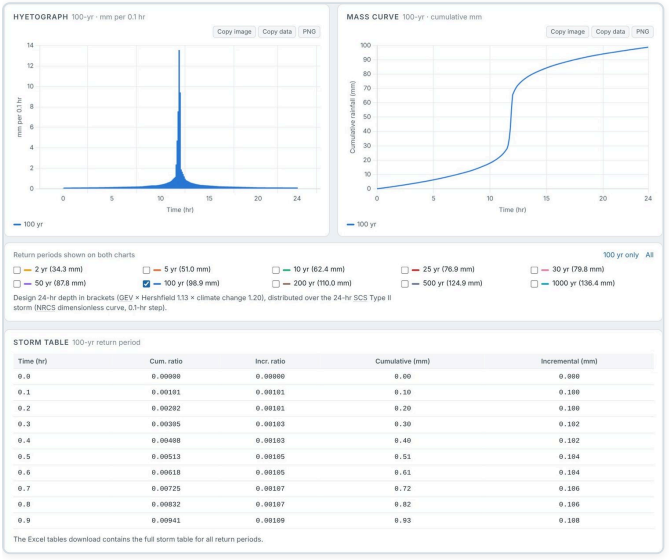


Figure 7.6 – Storm & DDF / IDF tab: design storm

One tab holds the design storm (top) and the DDF / IDF tables and curves (below). The design 24-hr depth of each return period is shown in brackets next to its tick, e.g. *100 yr (98.9 mm)*, and the note under the ticks gives the distribution and the Hershfield and climate change factors. The ratio columns of the storm and DDF tables are shown too.

Hyetograph (mm per 0.1 h) and **mass curve** (cumulative mm). The ticks under the charts choose the return periods shown, each in its own colour (default: 100 yr only; *100 yr only · All* shortcuts). The **storm table** lists every 0.1-h step for one return period; when several are ticked, buttons above the table switch between them (all return periods are in the Excel tables). The table shows 10 rows and scrolls. A legend under the hyetograph and the mass curve names each return period (100 yr in blue, 2 yr in amber).

SCS DDF / IDF

For each duration d the depth is the **largest rainfall in any window of length d** within the design storm (moving-window maximum), as a ratio of the 24-hour depth R17 Tool:

$$P_d = \max_t[C(t + d) - C(t)] \times \text{Design P24}, \quad i_d = P_d / d \tag{7.2}$$

Bell's ratio method (24-h storm only; default or imported ratios, step 4)

$$P_d = \text{ratio}(d) \times \text{Design P24} \quad \text{R2} \tag{7.3}$$

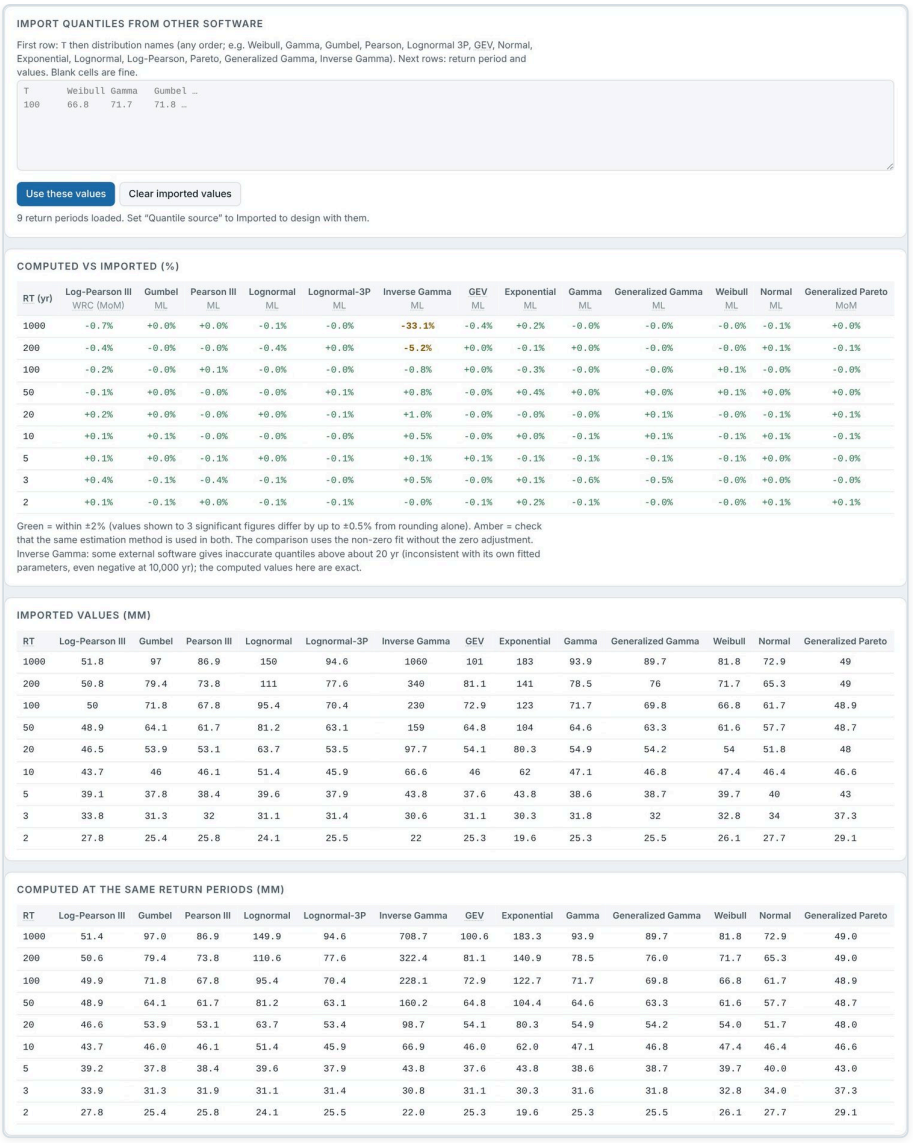
Charts: SCS DDF and IDF curves, the DDF and IDF curves of the ratio method (Bell's ratio by default), and below them the IDF comparison of the two methods at the chart return period (only when both methods are shown, step 4); all on log–log axes (duration 1 – 10,000 min) unless switched to linear (then 0 – 1,440 min).



Figure 7.7 – Storm & DDF / IDF tab: DDF / IDF part

7.6 Cross-check

Paste quantiles from other software (first row: T and distribution names). SORA shows the percentage difference for each distribution and return period (green within ±2 %, amber above). Set **Quantile source = Imported** in step 1 to design with the imported values. Differences usually come from rounding or a different estimation method – set the same methods on the Fitting tab (the list in the Method heading sets all at once) before comparing; some external programs give inaccurate Inverse-Gamma quantiles at long return periods. Tool



IMPORT QUANTILES FROM OTHER SOFTWARE

First row: T then distribution names (any order; e.g. Weibull, Gamma, Gumbel, Pearson, Lognormal 3P, GEV, Normal, Exponential, Lognormal, Log-Pearson, Pareto, Generalized Gamma, Inverse Gamma). Next rows: return period and values. Blank cells are fine.

```
T       Weibull Gamma   Gumbel …
100     66.8    71.7    71.8 …
```

Use these values | Clear imported values

9 return periods loaded. Set "Quantile source" to Imported to design with them.

COMPUTED VS IMPORTED (%)

RT (yr)	Log-Pearson III WRC (MoM)	Gumbel ML	Pearson III ML	Lognormal ML	Lognormal-3P ML	Inverse Gamma ML	GEV ML	Exponential ML	Gamma ML	Generalized Gamma ML	Weibull ML	Normal ML	Generalized Pareto MoM
1000	-0.7%	+0.0%	+0.0%	-0.1%	-0.0%	-33.1%	-0.4%	+0.2%	-0.0%	-0.0%	-0.0%	-0.1%	+0.0%
200	-0.4%	-0.0%	-0.0%	-0.4%	+0.0%	-5.2%	+0.0%	-0.1%	+0.0%	-0.0%	-0.0%	+0.1%	-0.1%
100	-0.2%	-0.0%	+0.1%	-0.0%	-0.0%	-0.8%	+0.0%	-0.3%	-0.0%	-0.0%	+0.1%	-0.0%	-0.0%
50	-0.1%	+0.0%	-0.0%	-0.0%	+0.1%	+0.8%	-0.0%	+0.4%	+0.0%	+0.0%	+0.1%	+0.0%	+0.0%
20	+0.2%	+0.0%	-0.0%	+0.0%	-0.1%	+1.0%	-0.0%	-0.0%	-0.0%	+0.1%	-0.0%	-0.1%	+0.1%
10	+0.1%	+0.1%	-0.0%	-0.0%	-0.0%	+0.5%	-0.0%	+0.0%	-0.1%	+0.1%	-0.1%	+0.1%	-0.1%
5	+0.1%	+0.0%	-0.1%	+0.0%	-0.1%	+0.1%	+0.1%	-0.1%	-0.1%	-0.1%	-0.1%	+0.0%	-0.0%
3	+0.4%	-0.1%	-0.4%	-0.1%	-0.0%	+0.5%	-0.0%	+0.1%	-0.6%	-0.5%	-0.0%	+0.0%	-0.0%
2	+0.1%	-0.1%	+0.0%	-0.1%	-0.1%	-0.0%	-0.1%	+0.2%	-0.1%	-0.0%	-0.0%	+0.1%	+0.1%

Green = within ±2% (values shown to 3 significant figures differ by up to ±0.5% from rounding alone). Amber = check that the same estimation method is used in both. The comparison uses the non-zero fit without the zero adjustment. Inverse Gamma: some external software gives inaccurate quantiles above about 20 yr (inconsistent with its own fitted parameters, even negative at 10,000 yr); the computed values here are exact.

IMPORTED VALUES (MM)

RT	Log-Pearson III	Gumbel	Pearson III	Lognormal	Lognormal-3P	Inverse Gamma	GEV	Exponential	Gamma	Generalized Gamma	Weibull	Normal	Generalized Pareto
1000	51.8	97	86.9	150	94.6	1060	101	183	93.9	89.7	81.8	72.9	49
200	50.8	79.4	73.8	111	77.6	340	81.1	141	78.5	76	71.7	65.3	49
100	50	71.8	67.8	95.4	70.4	230	72.9	123	71.7	69.8	66.8	61.7	48.9
50	48.9	64.1	61.7	81.2	63.1	159	64.8	104	64.6	63.3	61.6	57.7	48.7
20	46.5	53.9	53.1	63.7	53.5	97.7	54.1	80.3	54.9	54.2	54	51.8	48
10	43.7	46	46.1	51.4	45.9	66.6	46	62	47.1	46.8	47.4	46.4	46.6
5	39.1	37.8	38.4	39.6	37.9	43.8	37.6	43.8	38.6	38.7	39.7	40	43
3	33.8	31.3	32	31.1	31.4	30.6	31.1	30.3	31.8	32	32.8	34	37.3
2	27.8	25.4	25.8	24.1	25.5	22	25.3	19.6	25.3	25.5	26.1	27.7	29.1

COMPUTED AT THE SAME RETURN PERIODS (MM)

RT	Log-Pearson III	Gumbel	Pearson III	Lognormal	Lognormal-3P	Inverse Gamma	GEV	Exponential	Gamma	Generalized Gamma	Weibull	Normal	Generalized Pareto
1000	51.4	97.0	86.9	149.9	94.6	708.7	100.6	183.3	93.9	89.7	81.8	72.9	49.0
200	50.6	79.4	73.8	110.6	77.6	322.4	81.1	140.9	78.5	76.0	71.7	65.3	49.0
100	49.9	71.8	67.8	95.4	70.4	228.1	72.9	122.7	71.7	69.8	66.8	61.7	48.9
50	48.9	64.1	61.7	81.2	63.1	160.2	64.8	104.4	64.6	63.3	61.6	57.7	48.7
20	46.6	53.9	53.1	63.7	53.4	98.7	54.1	80.3	54.9	54.2	54.0	51.7	48.0
10	43.7	46.0	46.1	51.4	45.9	66.9	46.0	62.0	47.1	46.8	47.4	46.4	46.6
5	39.2	37.8	38.4	39.6	37.9	43.8	37.6	43.8	38.6	38.7	39.7	40.0	43.0
3	33.9	31.3	31.9	31.1	31.4	30.8	31.1	30.3	31.6	31.8	32.8	34.0	37.3
2	27.8	25.4	25.8	24.1	25.5	22.0	25.3	19.6	25.3	25.5	26.1	27.7	29.1

Figure 7.8 – Cross-check of the example gauge against quantiles imported from external software; the example loads with the same methods as that software (ML; Log-Pearson III WRC; Generalized Pareto MoM)

8 Checks, projects and exports

8.1 Automatic checks

Check	Meaning / action
Rainfall data present	Data loaded and read.
At least 20 years used	OK ≥ 20 years · REVIEW 10–19 · CHECK < 10. Shorter records give unreliable 100-yr and longer estimates R21 R6.
No manually entered maxima	REVIEW when any annual maximum was changed or added by hand – document its source.
No gap (missing) years	Lists missing calendar years (skipped, not zero).
No low outliers (MGBT, 17C)	Low outliers found; censored if MGBT is On R4.
Zero annual maxima handled	States how zero years were treated (Chapter 4.2).
No suspect years (max = 0 / incomplete)	Amber years: maximum = 0 or incomplete sub-daily year – check the wet season.
Hershfield factor (short-duration data)	REVIEW when a factor other than 1.00 is used with sub-daily data (Chapter 4.1).
No outliers flagged (Z-score)	High values above the Z limit – verify they are real.
All chosen methods available	Lists distributions where ML failed and MoM was used, and a Generalized Gamma ML fit stopped at its search limit (≈ Lognormal).
Return periods ascending	Table order valid.
Selected distribution valid · Imported values available (if used)	Selection and imported quantiles usable.
Design depths physically plausible	Only when a design 24-hr depth exceeds the world-record 24-hour rainfall (≈ 1,825 mm, La Réunion 1966): the fitted tail is unrealistic – review the distribution, the units and the data.
SCS storm totals = design depths · SCS ratio ends at 1.0 · Custom storm valid	Storm mass balance and curve integrity.
Custom storm: IDF intensity falls with duration	REVIEW when a custom storm gives a rainfall intensity that rises again at a longer duration (an unusually uniform pattern) – check the pasted curve. A custom storm that cannot be used falls back to SCS Type II with a CHECK.
Bell's ratio = 1.0 at 24 hr	Bell table integrity (24-h storm only).
Same results as when saved	Only for a gauge opened from a project saved by an earlier SORA version: REVIEW when the same inputs now give a different selected distribution or design P24 (≥ 0.05 mm); the note gives the saved version and what changed. Review the gauge, decide, then save the project again. After you change any input the check is dropped – it is then a new analysis.

Status: OK REVIEW engineering judgement needed · CHECK something is wrong · OFF not applicable. The gauge dot takes the worst status.

8.2 Projects and exports

Project files (top bar)

Button	What it does
New	Starts an empty project with one blank gauge. Unsaved changes are offered for saving first (*Save / Don't save / Cancel*).
Open...	Select one or several .sora files (project files of earlier versions, *.json*, open too). In Windows a .sora file shows the SORA icon and opens in SORA with a double-click; .sora files can also be dropped on the data box. Each gauge opens with the settings saved in the project (tests, zeros and outliers, storm, return periods, methods, notes), so the analysis is reproduced as it was saved; only the fixed settings (95 % limits, best estimate, Z = 3, DDF durations) are applied. If no gauge holds data, the opened gauges replace the blank one; otherwise they are added (duplicate names get "(2)", "(3)" ...; a project opened twice gives independent copies; a name already in use asks Update / Replace / Keep both, as in Chapter 3).
Save · Save as...	Both open the Windows *Save as* window. With one gauge: one **.sora** project file. With two or more gauges SORA asks: *One file with all gauges* (.sora) or *One file per gauge* (a ZIP with one .sora per gauge, named after the gauge). The file holds the data and settings, and also each gauge's key results (selected distribution, design P24), so a later version of SORA can say if the same inputs give different results.

Export ▾

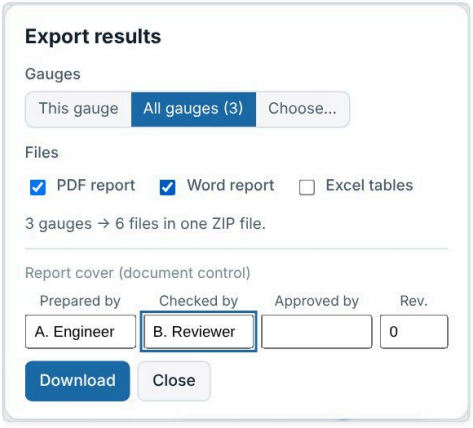


Figure 8.1 – Export results (three gauges)

Gauges (two or more gauges with data): *This gauge*, *All gauges (n)* or *Choose...* (a list with search, All and None).

Files: **PDF report** (ready to issue), **Word report** (editable, same content: contents, document control, figures and tables), **Excel tables** (summary, annual maxima, distributions, quantiles, goodness of fit, full SCS storm, DDF/IDF; always complete). The line below says how many files will be made.

Report cover (document control): *Prepared by*, *Checked by*, *Approved by* and *Rev.* (default 0) are printed in the document-control table of the reports. They are typed for each export and not kept.

Click **Download**: the *Save as* window comes first, then the files are made (a progress bar shows the work). One gauge and one file type gives that file; otherwise one ZIP (split into parts of 100 gauges, each part asking where to save it). A gauge without results is listed with the reason. Notes are never exported.

Report contents

Section	Content
Document control, summary of results, introduction	Cover details, design values, selected distribution, data used.
Rainfall data, annual-maximum chart and table	Full statistics.

Data screening	Outliers, MGBT, zeros, missing years.
Methodology	Methods and parameter table.
Fitting results	All quantiles and goodness of fit.
Selection of the design distribution	Justification, top 5 with quantiles.
Design 24-hour rainfall	With 95 % confidence interval.
Design storm, DDF/IDF (SCS and ratio method)	Tables and charts.
Cross-check (if imported values)	Computed vs imported quantiles.
Assumptions and limitations, references	As used for the gauge.

The justification paragraph explains why the distribution was selected (its ranks, the BIC preference, the spread of the top five at the chart return period, any upper bound and the record-length limit on extrapolation). Wide tables are split into parts automatically. Tool

9 Several gauges, notes, dark mode and the AI assistant

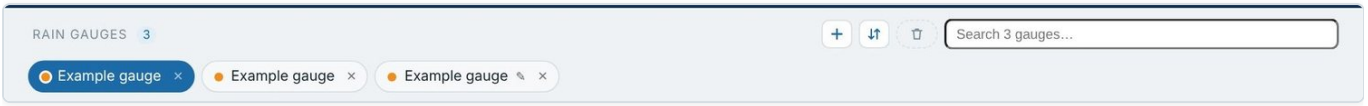


Figure 9.1 – The rain gauges bar with three gauges

Element	Use
Gauge chips	Click to open; double-click to rename; × twice to remove (Ctrl+Z restores it). Dot = status; "✎" = the gauge has notes (hover to read them). The number of gauges is shown next to *Rain gauges*.
+ (Add a gauge)	Empty gauge with the current settings; step 1 opens for its data.
⇅ (Order)	Order of the chips – click to switch: *Newest first* (default) → *A → Z* → *Z → A* → back. Names sort naturally (Gauge 2 before Gauge 10). Only the display order changes; the choice is kept on this computer.
Bin	Removes every gauge (click twice); with one gauge it clears that gauge. Ctrl+Z restores them while SORA stays open – save the project to be safe.
Search box	Type part of a name: only the matching gauges are shown; Enter opens the first, Esc clears the search.

Good to know

- Up to 1,000 gauges were tested: reading ≈ 3 s, analysis ≈ 30 s (in the background while you work), reload < 1 s.
- Keyboard: **Alt+←/→** switches to the previous/next gauge; **Ctrl+Z / Ctrl+Y** undo and redo; **Esc** closes menus and the gauge search (when its box has focus). ⇅, the search box and the *Apply … to all gauges* buttons appear when there are two or more gauges.
- **Save your work as project files.** SORA starts empty each time; closing it with unsaved changes asks *Save / Don't save / Cancel* (*Save* closes SORA only after the project file is saved).

One rainfall reference for the team

Analyse every gauge once, with the distributions and settings your team has agreed, and save the gauges as .sora project files on a shared drive. Any engineer then opens a gauge and gets the same data, settings, selected distribution and design rainfall, with the notes of the engineer who prepared it (Chapter 9.1) – no re-analysis and no different answers for the same gauge. Tool

9.1 Notes for the next engineer

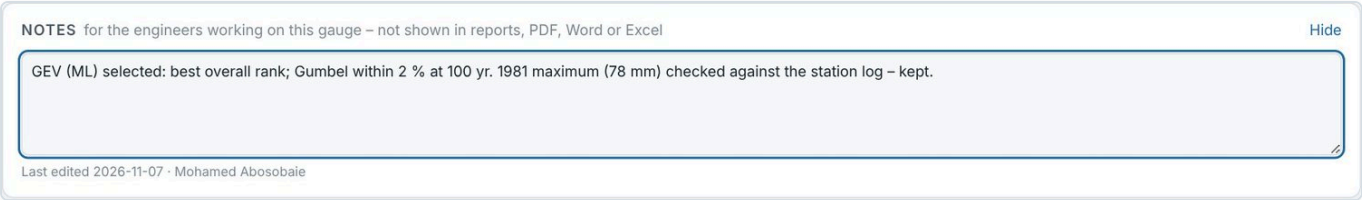


Figure 9.2 – Notes of a gauge

✎ **Notes** (next to *Export* ▾) opens a notes box for the open gauge – e.g. why a distribution and method were chosen, data issues, years checked with the client, open questions. The notes are saved with the gauge and in project files, so the next engineer who opens the project reads them: a gauge with notes shows "✎" on its chip (hover to read) and its notes box opens automatically. The date of the last edit is shown under the box; *Hide* folds it away.

Notes are for the engineers only: they never appear in any exported file (PDF, Word or Excel). Tool

9.2 Dark mode

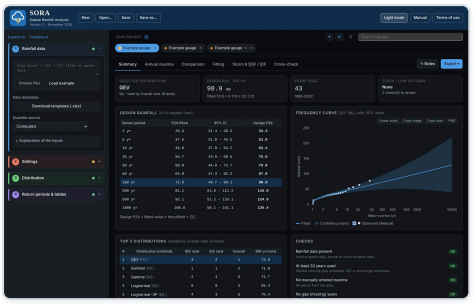


Figure 9.3 – Dark mode

Dark mode in the top bar switches SORA to dark panels with bright chart colours; the button then reads **Light mode** to switch back. The choice is kept on this computer and applies to every gauge.

Calculations, results, checks and reports are identical in both modes – Word and PDF reports always use the standard SORA colours on a white page; only the screen changes. Tool

9.3 AI assistant

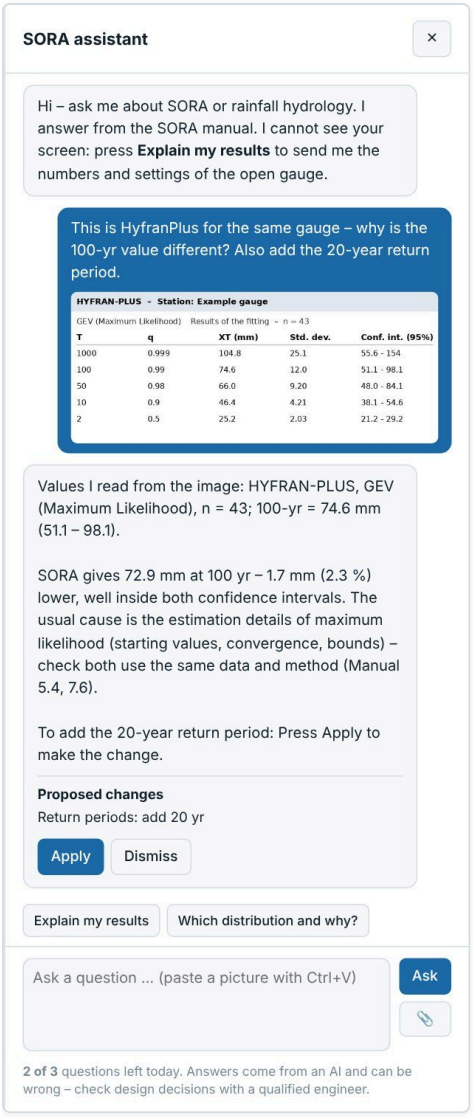


Figure 9.4 – The SORA assistant: a picture compared and input changes proposed

Assistant in the top bar opens a side panel where you can ask questions about SORA and rainfall hydrology. Answers come from an AI service through the developer's server and are based on this manual: for questions about SORA the assistant names the manual section it used, and when the manual does not cover something it says so.

- **It cannot see your screen.** A typed question sends only the question and the last few messages. **Explain my results** and **Which distribution and why?** also send the numbers and step settings of the open gauge (annual maxima, settings, fitted and design values, ranks, test results and checks) – never file, gauge or project names.
- **Pictures** – click 📎, paste a screenshot into the question box (Ctrl+V) or drop it on the panel, e.g. results from other software or your own calculation, and ask why they differ. The picture is read first and the answer starts with *Values I read from the image* – check them before trusting the comparison. The open gauge's numbers are sent with the picture so they can be compared; pictures are not kept. For exact quantile checks, the Cross-check tab (Chapter 7.6) is more reliable than a picture.
- **Changing inputs** – ask, for example, “set the climate change factor to 1.1”, “use GEV for all gauges” or “add the 20-year return period”. The assistant lists the proposed changes as *now → new* with **Apply** and **Dismiss**; nothing changes until you press Apply, and **Ctrl+Z** undoes it. It can change the step settings only (factors, zeros & outliers, SCS storm type and duration, fitting method, selection, tests, distributions used, quantile source, return periods, DDF/IDF methods and *Apply ... to all gauges*) – never your data, files or exports.
- **3 questions per computer per day**; pressing Apply is not a question. The line under the box shows how many are left. The developer can change the limit, switch the assistant off, or turn it off for a computer.
- It needs the internet. Answers come from an AI and can be wrong – they are not engineering advice; check design decisions with a qualified engineer and your local standards.

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Tool marks conventions chosen in SORA.

Appendix A Distributions and formulas

F = non-exceedance probability (1 − 1/T), z = standard normal variate of F. Parameter names are those shown on the Fitting tab. R18 R11 R5

Distribution	Parameters	Quantile x_T	Methods
Normal	μ, σ	$\mu + \sigma z$	ML, MoM, LM
Lognormal	μ_y, σ_y (natural logs)	$\exp(\mu_y + \sigma_y z)$	ML, MoM, LM
Lognormal-3P	c, μ_y, σ_y (sign s)	$c + s \cdot \exp(\mu_y + s\,\sigma_y z)$	ML, MoM, LM
Exponential	x_o, β	$x_o - \beta \ln(1 - F)$	ML, MoM, LM
Gumbel (EV1)	u, α	$u - \alpha \ln(-\ln F)$	ML (bias-corrected), MoM, LM
GEV	ξ, α, k (Hosking sign: k > 0 bounded above)	$\xi + (\alpha/k)[1 - (-\ln F)^k]$	ML, MoM, LM
Weibull	k, λ	$\lambda[-\ln(1 - F)]^{1/k}$	ML, MoM, LM
Gamma	a (shape), b (scale)	$G^{-1}(F; a, b)$	ML, MoM, LM
Pearson III	c, a, b (sign s)	$c + s \cdot G^{-1}(F \text{ or } 1 - F; a, b)$	ML, MoM, LM
Log-Pearson III	c, a, b on ln x	exp[Pearson III quantile of ln x]	WRC / EMA
Generalized Pareto	α, k (location 0)	$(\alpha/k)[1 - (1 - F)^k]$	MoM, ML, LM
Generalized Gamma	a, c (power), b	$b \cdot [G^{-1}(F; a, 1)]^{1/c}$	ML
Inverse Gamma	α, λ	$\lambda / G^{-1}(1 - F; \alpha, 1)$	ML, MoM

G^{-1} = inverse gamma CDF. For Pearson III / LP3 with very small skew (gamma shape > 2500) the Wilson–Hilferty approximation is used R28. Rainfall quantiles below zero are set to 0.

Key L-moment and moment relations

ℓ_1 = mean, $\ell_2 = 2b_1 - b_0$, $\tau_3 = (6b_2 - 6b_1 + b_0)/\ell_2$ (b_r = probability-weighted moments) R8

Gumbel LM: $\alpha = \ell_2 / \ln 2$, $u = \ell_1 - 0.5772\,\alpha$ · Weibull LM: $k = -\ln 2 / \ln(1 - \ell_2/\ell_1)$ · GPD LM: $k = \ell_1/\ell_2 - 2$, $\alpha = (1 + k)\,\ell_1$ R11

GEV LM: $c = 2/(3 + \tau_3) - \ln 2/\ln 3$, $k \approx 7.859c + 2.9554c^2$ (refined by Newton) R9

Gumbel MoM: $\alpha = \sqrt{6}\,s / \pi$, $u = \bar{x} - 0.5772\,\alpha$ R5

Appendix B Verification of SORA

SORA was verified in October 2026 on **100,920 rainfall records** (5 – 150 years) drawn from 14 parent distributions and resampled from 60 real gauges, with random settings. The records include zeros, gap years, high and low outliers, ties, mixed units, very short and very long, constant and all-zero series.

Check	Size	Result
Calculation rules: quantiles rise with return period, limits around the estimate, storm mass balance, DDF increasing / IDF decreasing, zero and gap handling, unit and order independence, repeatability	100,920 records	no violation
Quantile functions vs SciPy	≈ 1.27 million quantiles	identical (< 0.002 %)
Log-likelihood (AIC/BIC), Anderson–Darling, PPCC vs SciPy	2,146 records × 13 distributions	identical (< 1e-6)
Method-of-moments fits reproduce sample moments	32,600 fits	pass
L-moment fits vs *lmoments3*	27,400 fits	pass
Maximum-likelihood optimum (independent optimiser)	2,146 records × 13 distributions	never improved (regular fits; Gumbel excluded – its ML is bias-corrected by design; Normal and Lognormal use the n − 1 standard deviation, within 0.02 of the maximum log-likelihood)
Confidence limits vs independent numerical Hessian	≈ 8,000 values	within 2 % (regular cases)
File reading (daily, Year/Day table, monthly table, annual, sub-daily incl. rows out of order, repeated and missing time steps; date formats, delimiters, CSV and Excel; rain-gauge report .txt files checked against their own TOTALS lines)	3,500 generated files and the real gauge files	all read correctly
Real files kept as a fixed reference set (daily, Year/Day, annual and short-duration gauges, and workbooks): layout, annual maxima, selected distribution and design P24 must be identical before and after every change; any difference is explained before release	33 files, 34 gauges	identical
Reports: values in PDF, Word and Excel vs the screen	250 gauges × 3 formats	identical
Interface: automated checks and random user actions (4 screen widths, both themes)	41 checks, 1,200 random actions	no error

Appendix C Limitations and good practice

- **Single-station analysis.** Results represent the gauge; no areal reduction is applied. For catchments, apply an areal reduction factor where required R21.
- **Record length.** Estimates beyond about twice the record length are extrapolations; read them with their confidence limits. SORA checks for at least 20 years; prefer 20–30 years or more for 100-year values R21 R6.
- **Data quality first.** Review amber/red calendar years and flagged outliers before trusting the fit.
- **Distribution choice.** When several well-ranked distributions differ strongly at the design return period, look at the probability plot and consider the more conservative curve or a regional analysis R11.
- **Hershfield and climate factors** are uniform multipliers; use the values required by your project or authority and document them in step 2.
- **Temporal pattern.** SCS curves were developed for the USA; use a local design storm (Custom) where one exists R17.
- **Bell's ratios** use SORA's built-in default set; replace with regional ratios where available R2.
- **Generalized Gamma** ML can sit on a flat likelihood near the lognormal limit; long-return-period values may then vary by 1–3 %.
- **Saving.** SORA starts empty each time; keep project files (.sora).

Appendix D Glossary

Term	Meaning
AMS	Annual maximum series – the largest value of each year.
Return period T	Average interval (years) between exceedances; T = 1 / annual exceedance probability.
P24	24-hour rainfall depth (mm).
Design P24	Quantile × Hershfield × climate factor (Eq. 4.1).
Hershfield factor	Converts fixed-interval daily maxima to true 24-hour maxima (1.13 daily; 1.00 by default for sub-daily data).
Quantile	Value with a given non-exceedance probability.
ML · MoM · LM · WRC · EMA	Maximum likelihood · method of moments · L-moments · moments of the logs (Bulletin 17) · Expected Moments Algorithm.
MGBT · PILF	Multiple Grubbs–Beck test · potentially influential low flood (low outlier).
Censoring	Using a value only as “below threshold”, not its exact size.
Conditional adjustment	Combining the probability of zero/censored years with the fitted distribution (Eq. 6.1).
GOF	Goodness of fit (Chi-squared, RMSE, IOA, NSE).
AIC · BIC	Information criteria balancing fit and number of parameters.
Confidence limits	Range likely (95 %) to contain the true quantile.
Hyetograph · mass curve	Rainfall per time step · cumulative rainfall over the storm.
DDF · IDF	Depth– · intensity–duration–frequency relationships.
Bell's ratio	Ratio of a short-duration depth to the 24-hour depth.

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